

## UNIT-II

### Semiconductor Physics and Devices

#### Introduction:

The materials which allow electricity partially to pass through them are known as semiconductors. From last two decades the use of solid state devices has become prominent. Most of these solid state devices use Semi conductor crystals. From the point of band theory of electrical conductivity, semiconductors differ from conductors and insulators. In metals addition of impurities and rise of temperature result in decrease in the electrical conductivity where as in semi conductors the electrical conductivity increases in contrary to conductors. Germanium and Silicon are most important semiconductors which are widely used in the manufacturing of diodes and transistors.

#### 2.1 Types of Semi Conductors:

Semiconductors are mainly classified into two types.

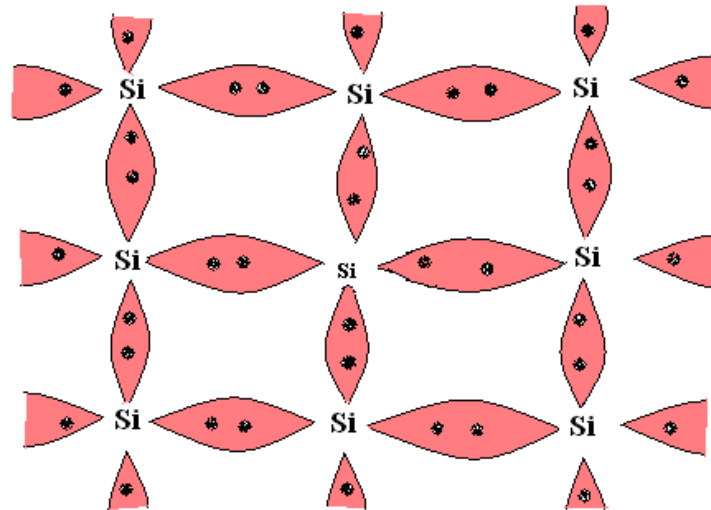
1. Intrinsic (or) Pure Semi conductor
2. Extrinsic (or) Impure semiconductor.

#### Intrinsic Semiconductor:

A semiconductor in its pure form is known as intrinsic semiconductor.

- The Silicon crystal which contains all of Si atoms is known as intrinsic semiconductor.
- In the Si (or) Ge the number of valence electrons is four. Each atom will complete '4' covalent bonds with four similar neighboring atoms.
- At the room temperature due to the thermal agitation some electrons in the valence band are shifted into the conduction band. Then electron vacancies created in the valence band. These are known as holes.
- The number of holes in the valence band and number of electrons in conduction band are equal.
- The average energy of holes and the electrons are represented by the Fermi energy level. It lies at the middle of forbidden gap.

Ex: - Ge, Si.



**Fig:** Pure Silicon

### Extrinsic Semiconductor:

A doped semiconductor is called extrinsic semiconductor. The process of adding impurities to a substance is called doping. On basis of doping Extrinsic semiconductors are again divided into two types.

1. N-type extrinsic semiconductor
2. P- type extrinsic semiconductor

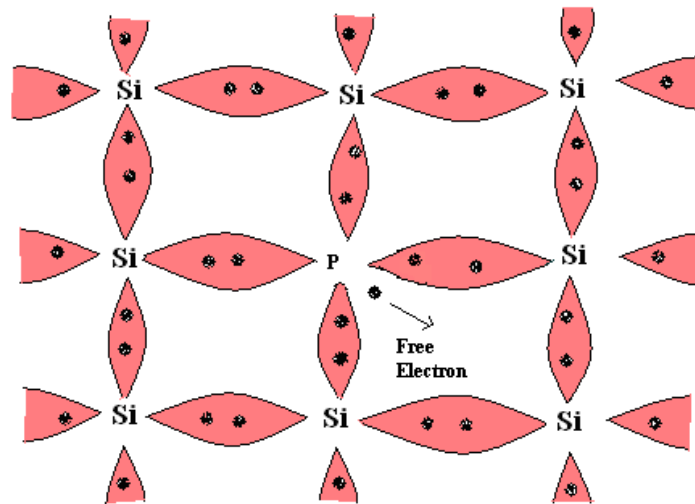
#### 1. N-type Extrinsic Semiconductor:

When we add the pentavalent impurity to a pure semiconductor then it is called N-type extrinsic semiconductor.

- This semiconductor is generated by doping pentavalent impurity to the intrinsic semiconductor.
- The phosphorous (p) atom is substituting in the place of 'Si' atom then it will complete four covalent bonds with four neighboring Si atoms. But one electron is left free without forming the bond. This electron goes into the conduction band without creating hole in the valence band.
- A small quantity of pentavalent impurity is added to the crystal then large number of electrons is produced in conduction band. Due to thermal agitation some electrons in the valence band are shifted into the conduction band by creating the holes in the valence band.
- The impurity atom is donating an electron to the conduction band. So this impurity is known as donor impurity.
- The majority charge carriers are electrons and the minority carriers are holes.

- The majority carriers are the negative charge carriers. So this semiconductor is known as N-type extrinsic semiconductor.
- The Fermi energy level is shifted towards the conduction band.

Ex: Pentavalent impurities are P, As, etc.



**Fig:** Pentavalent impurity (P) doped into intrinsic semiconductor (Si)

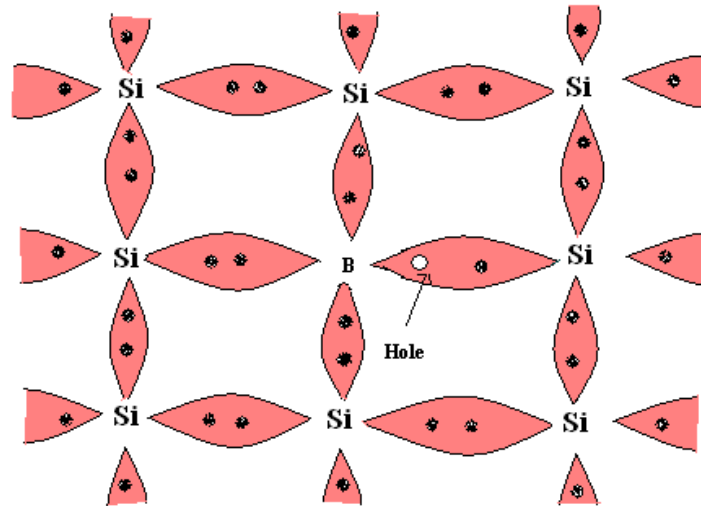
## 2. P-type Extrinsic Semiconductor:

When we add trivalent impurity to a pure semiconductor then it is called a P-type extrinsic semiconductor.

- This semiconductor is generated by doping trivalent impurity to the intrinsic semiconductor.
- If an 'Al' atom is substituted in place of 'Si' atom then it will complete three covalent bonds with three neighboring 'Si' atoms. But one more electron should be required to complete the fourth covalent bond. In the absence of fourth electron a hole is generated which lies in valence band. In this process no electron is shifted to the conduction band.
- A small quantity trivalent impurity is added to the crystal then large number of holes in the valence band. Due to thermal agitation some more electrons in valence band are shifted to conduction band by creating the holes in valence band.
- The impurity atom is in a position to get an electron. So this impurity atom is known as acceptor impurity..
- In this semiconductor majority carriers are holes and minority carriers are electrons.
- The majority carriers have positive charge, so this semi conductor is known as P-type extrinsic semiconductor.

➤ The Fermi energy level shifted towards the valence band.

Ex: - B, Al, Ga, etc.



**Fig:** Trivalent impurity (B) doped into intrinsic semiconductor (Si)

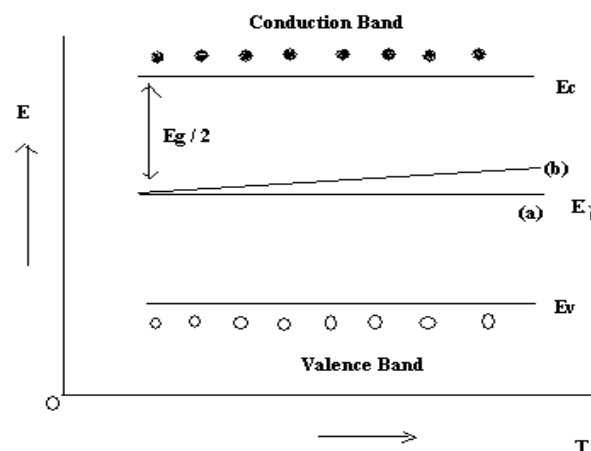
## 2.2 Carrier Concentration in Semiconductors

The number of charge carriers per unit volume is defined as carrier concentration.

Units: Number of electrons/m<sup>3</sup>.

### Intrinsic Carrier Concentrations:

In intrinsic semi conductors, as the temperature is increased electron-hole passes will be generated. Hence the electron concentration,  $n_i$ , in the conduction band will be equal to the hole concentration  $p_i$  in the valence band.



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Fig: Fermi level at (a)  $T = 0$  (b) As temperature increases  $m_h^* > m_e^*$

Let  $E_i$  be the Fermi level of the intrinsic semiconductor in equilibrium, then the number of electrons per unit volume in the conduction band.

$$n_i = 2 \left( \frac{2m_e \pi k_B T}{h^2} \right)^{\frac{3}{2}} \exp \left( \frac{E_i - E_c}{k_B T} \right)$$

And the number of holes per unit volume in the valence band

$$p_i = 2 \left( \frac{2m_h \pi k_B T}{h^2} \right)^{\frac{3}{2}} \exp \left( \frac{E_v - E_i}{k_B T} \right)$$

But in intrinsic semiconductors  $n_i = p_i$

$$\text{Hence, } 2 \left( \frac{2m_e \pi k_B T}{h^2} \right)^{\frac{3}{2}} \exp \left( \frac{E_i - E_c}{k_B T} \right) = 2 \left( \frac{2m_h \pi k_B T}{h^2} \right)^{\frac{3}{2}} \exp \left( \frac{E_v - E_i}{k_B T} \right)$$

$$m_e^{\frac{3}{2}} \exp \left( \frac{E_i - E_c}{k_B T} \right) = m_h^{\frac{3}{2}} \exp \left( \frac{E_v - E_i}{k_B T} \right)$$

$$\exp \left( \frac{E_i - E_c - E_v + E_i}{k_B T} \right) = \left( \frac{m_h}{m_e} \right)^{\frac{3}{2}}$$

$$2E_i - E_c - E_v = K_B T \ln \left( \frac{m_h}{m_e} \right) K_B T$$

$$E_i = \frac{1}{2} (E_c + E_v) + \frac{3}{4} \ln \left( \frac{m_h}{m_e} \right) k_B T.$$

If the effective masses  $m_e = m_h$ , then

$$E_i = \frac{E_c + E_v}{2} [\because \ln 1 = 0]$$

**i.e.  $E_i$  lies mid way between  $E_c$  and  $E_v$ , which happens to be the centre of the bond gap at any temperature.**

Combining  $n_i$  and  $p_i$  values,

$$n_i p_i = 4 \left( \frac{2\pi m_e kT}{h^2} \right)^{\frac{3}{2}} \left[ \frac{2\pi m_h k_B T}{h^2} \right]^{\frac{3}{2}} \exp \left( \frac{E_i - E_c}{KT} \right) \exp \left( \frac{E_v - E_i}{KT} \right)$$

$$n_i^2 = 4 \left( \frac{2\pi k}{h^2} \right)^3 (m_e m_h)^{\frac{3}{2}} T^3 \exp \left( \frac{E_v - E_c}{KT} \right) (\because n_i = p_i)$$

$$n^2 = AT^3 \exp \left( \frac{-E_g}{KT} \right)$$

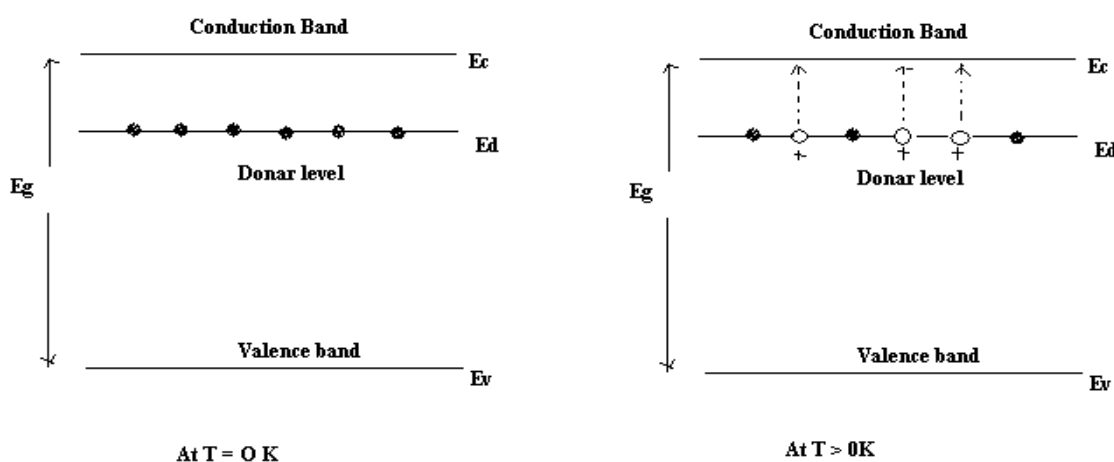
Where  $A = 4 \left( \frac{2\pi k}{h} \right)^3 (m_h m_e)^{\frac{3}{2}}$  a constant value of semiconductor and  $E_g = E_c - E_v$  is energy gap of the semiconductor.

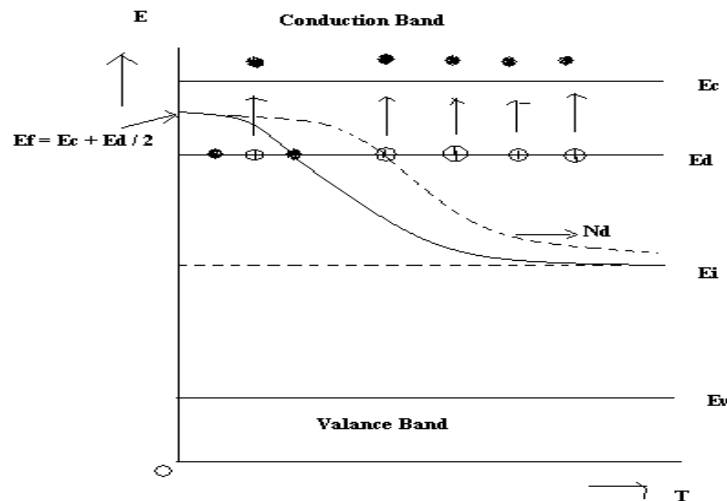
It is to be noted that A and  $E_g$  are constants for a given semiconductor and  $n_i$  is called the intrinsic concentration which indicates thermally generated electrons and holes. It is a strong function of temperature 'T'.

## 2.3 Electron Concentration in Extrinsic Semiconductor

### 1. Carrier Concentration in N-Type Semiconductors:

The energy level diagram of a N-type semiconductor shown in figure





**Fig:** Variation of Fermi Level with Temperature

$N_d$  is the donor concentration and  $E_d$  is the donor energy level. At very low temperatures all donor levels are filled with electrons. With increase of temperature more and more donor atoms get ionized and the density of electrons in the conduction band increases.

Density of electrons in conduction band is given by

$$n = 2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{\frac{3}{2}} \exp \left( \frac{E_F - E_C}{KT} \right) \text{---(1)}$$

If we assume that  $E_F$  lies more than a few  $KT$  above the donor level then the density of ionized donors is given by

$$N_d [1 - F(E_d)] \approx N_d \exp \left( \frac{E_d - E_F}{KT} \right)$$

At very low temperatures, when electron-hole pairs not generated due to breaking of covalent bonds. The number of electrons in the conduction band must be equal to the number of ionized donors.

$$\text{i.e. } 2 \left( \frac{2\pi m_e^* KT}{h^2} \right)^{\frac{3}{2}} \exp \left( \frac{E_F - E_C}{KT} \right) = N_d \exp \left( \frac{E_d - E_F}{KT} \right)$$

Taking logarithm and rearranging we get

$$\left(\frac{E_F - E_C}{KT}\right) - \left(\frac{E_d - E_F}{KT}\right) = \log N_d - \log 2 \left(\frac{2\pi m_e^* KT}{h^2}\right)^{\frac{3}{2}}$$

$$2E_F - (E_d + E_C) = KT \log \frac{N_d}{2 \left(\frac{2\pi m_e^* KT}{h^2}\right)^{\frac{3}{2}}}$$

$$E_F = \left(\frac{E_d + E_c}{2}\right) + \frac{kT}{2} \log \frac{N_d}{2 \left(\frac{2\pi m_e^* kT}{h^2}\right)^{\frac{3}{2}}} \text{-----(2)}$$

$$\text{At } 0 \text{ K. } E_F = \frac{E_d + E_c}{2}$$

i.e. at 0 k, Fermi level lies exactly at the middle of the donor level  $E_d$  and the bottom of the conduction band  $E_c$

### Density of Electrons in the Conduction Band:

The density of electrons in the conduction band is given by

$$n = 2 \left(\frac{2\pi m_e^* KT}{h^2}\right)^{\frac{3}{2}} \exp\left(\frac{E_F - E_C}{KT}\right) \text{-----(3)}$$

Substitute ' $E_F$ ' value in equation (2) in the above equation.

$$\exp\left(\frac{E_F - E_c}{KT}\right) = \exp\left\{\left(\frac{E_d - E_c}{2KT}\right) + \frac{1}{2} \log \frac{N_d}{2 \left(\frac{2\pi m_e^* KT}{h^2}\right)^{\frac{3}{2}}}\right\}$$

$$\text{Exp}\left(\frac{E_F - E_C}{KT}\right) = \exp\left\{\left(\frac{E_d - E_c}{2KT}\right) + \log \frac{N_d^{\frac{1}{2}}}{\left[2 \left(\frac{2\pi m_e^* KT}{h^2}\right)^{\frac{3}{2}}\right]^{\frac{1}{2}}}\right\} \left(\because \frac{1}{2} \log x = \log x^{\frac{1}{2}}\right)$$



$$= \left[ \exp\left(\frac{E_d - E_C}{2KT}\right) \right] \frac{N_d^{\frac{1}{2}}}{\left\{ 2 \left( \frac{2\pi m_e^* KT}{h^2} \right)^{\frac{3}{2}} \right\}^{\frac{1}{2}}} \left[ \because \exp(a+b) = e^a \cdot e^b \right]$$

Hence

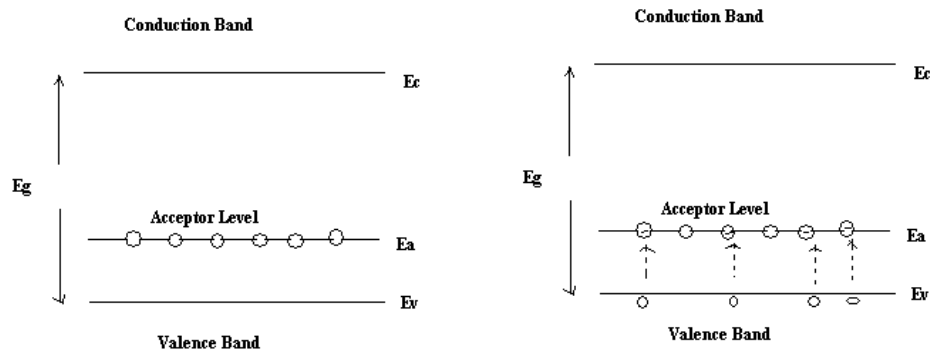
$$n = 2 \left( \frac{2\pi m_e^* KT}{h^2} \right)^{\frac{3}{2}} \frac{N_d^{\frac{1}{2}}}{\left[ 2 \left( \frac{2\pi m_e^* KT}{h^2} \right)^{\frac{3}{2}} \right]^{\frac{1}{2}}} \exp\left(\frac{E_d - E_C}{2KT}\right)$$

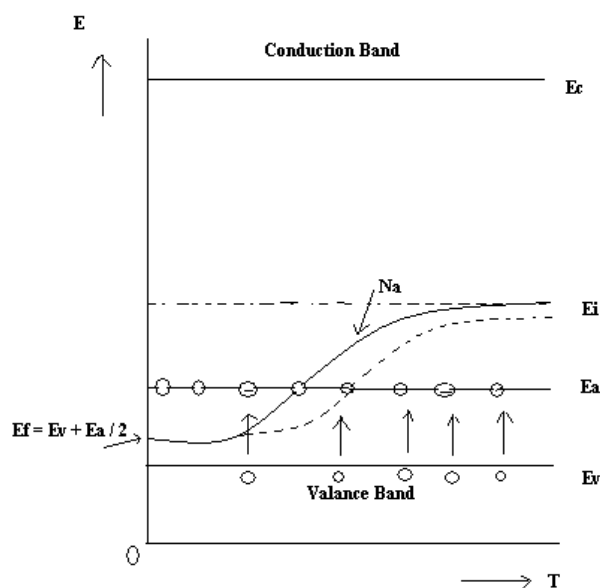
$$n = (2N_d)^{\frac{1}{2}} \left( \frac{2\pi m_e^* KT}{h^2} \right)^{\frac{3}{4}} \exp\left(\frac{E_d - E_C}{KT}\right)$$

Thus we find that the density of electrons in the conduction band is proportional to the square root of the donor concentration at moderately low temperatures.

## 2. Carrier Concentration in P-type Semiconductor:

The energy level diagram of a P-type semiconductor is as shown in figure below.





**Fig:** Variation of Fermi Level with Temperature

$N_a$  is the acceptor concentration i.e. the number of acceptor atoms per unit volume of the material and  $E_a$  is the acceptor energy level.

At very low temperatures all the acceptor levels are empty. With increase of temperature acceptor atoms get ionized i.e. the electrons move from valence band and occupy the vacant sites in the acceptor energy level thereby leaving holes in the valence band.

Density of holes in the valence band is given by

$$p = 2 \left( \frac{2\pi m_h^* K T}{h^2} \right)^{\frac{3}{2}} \exp \left( \frac{E_v - E_F}{K T} \right)$$

Since  $E_F$  lies below acceptor level, the density of ionized acceptor is given by

$$N_a F(E_a) = N_a \exp \left( \frac{E_F - E_a}{K T} \right)$$

Since the density of holes in the valence band is equal to the density of ionized acceptors

$$2 \left( \frac{2\pi m_h^* KT}{h^2} \right)^{\frac{3}{2}} \exp \left( \frac{E_V - E_F}{KT} \right) = N_a \exp \left( \frac{E_F - E_a}{KT} \right)$$

$$\text{i.e. } \exp \left( \frac{E_V + E_a - 2E_F}{KT} \right) = \frac{N_a}{2 \left( \frac{2\pi m_h^* KT}{h^2} \right)^{\frac{3}{2}}}$$

Taking logarithm

$$\left( \frac{E_V + E_a - 2E_F}{KT} \right) = \log \frac{N_a}{2 \left( \frac{2\pi m_h^* KT}{h^2} \right)^{\frac{3}{2}}}$$

$$\text{i.e. } E_F = \frac{E_V + E_a}{2} - \frac{KT}{2} \log \frac{N_a}{2 \left( \frac{2\pi m_h^* KT}{h^2} \right)^{\frac{3}{2}}} \text{-----(1)}$$

$$\text{AT O K } E_F = \frac{E_V + E_d}{2}$$

At Ok, Fermi level lies exactly at the middle of the acceptor level and the top of the valence band as shown in above fig

### Density of Holes in Valence Band:

The density of holes in the valence band is given by

$$p = 2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{\frac{3}{2}} \exp \left( \frac{E_V - E_F}{KT} \right) \text{-----(2)}$$

Substitute equation (1) value in equation (2)

$$\begin{aligned} \exp\left(\frac{E_V - E_F}{KT}\right) &= \exp\left\{\frac{E_V}{KT} - \frac{E_V + E_d}{2KT} + \frac{1}{2} \log \frac{N_a}{2\left(\frac{2\pi m_h^* KT}{h^2}\right)^{\frac{3}{2}}}\right\} \\ &= \exp\left\{\left(\frac{E_V - E_a}{2KT}\right) + \log \frac{N_a^{\frac{1}{2}}}{\left(2\left(\frac{2\pi m_h^* KT}{h^2}\right)^{\frac{3}{2}}\right)^{\frac{1}{2}}}\right\} \\ \text{Hence } p &= 2\left(\frac{2\pi m_h^* kT}{h^2}\right)^{\frac{3}{2}} \left\{\exp\left(\frac{E_V - E_a}{2KT}\right)\right\} \frac{N_a^{\frac{1}{2}}}{\left[2\left(\frac{2\pi m_h^* kT}{h^2}\right)^{\frac{3}{2}}\right]^{\frac{1}{2}}} \\ p &= (2N_a)^{\frac{1}{2}} \left(\frac{2\pi m_h^* KT}{h^2}\right)^{\frac{3}{4}} \exp\left(\frac{E_V - E_a}{2KT}\right). \end{aligned}$$

Thus we find that the density of holes in the valence band is proportional to the square root of the acceptor concentration at moderately low temperature.

## 2.4 Drift Current and Conductivity in Semiconductors

In the presence of electric field, the drift velocities  $V_{de}$  and  $V_{dh}$  carriers on the thermal velocities  $v_{te}$  and  $v_{th}$ . But the flow of charge carriers result in an electric current in the semiconductor crystal known as the drift currents. Let an electric field  $E$  be applied in the positive X-direction creating drift currents  $J_{nd}$  and  $J_{pd}$  of electrons and holes respectively.

In the absence of electric field, the drift velocities reduce to zero. Thus current density will also be zero. Consider the free electrons in a semiconductor moving with a uniform velocity  $v_{de}$ . In the negative X-direction due to the application of the electric field  $E$  in the positive X-direction. Suppose AB, a small rectangular box element of length  $v_{de}$  and unit sides of the square end faces A and B in the semiconductor shown in fig

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{Total change in the box AB}  $Q = \{\text{Volume of the box AB}\} \times \{\text{Density of Particles}\} \times \{\text{Charge on each particle}\}$

$$Q = (V_{de} \times 1 \times 1) \times n \times -q$$

$$\therefore Q = -qnV_{de} \text{ --- (1)}$$

The current density  $J_{nd}$  due to the free electrons at the face B will be

$$J = \frac{\text{charge}}{\text{time} \times \text{area}} \text{ --- (2)}$$

From (1) & (2) the current density of electrons  $J_{nd} = -qnV_{de} \text{ --- (3)}$

Similarly current density of holes,  $J_{pd} = qpV_{dh} \text{ --- (4)}$

But drift velocities in terms of mobilities are

$$V_{de} = -\mu_n E \text{ --- (5)}$$

$$V_{dp} = \mu_p E \text{ --- (6)}$$

Hence substituting equations (5) & (6) in (3) & (4)

$$J_{nd} = nq\mu_n E$$

$$J_{pd} = nq\mu_p E$$

Total current density due to both electrons and holes will be

$$J_d = J_{nd} + J_{pd}$$

$$J_d = (nq\mu_n + nq\mu_p)E$$

According to ohm's law, the current density

$$J_d = \sigma E$$

$$\sigma = nq\mu_n + nq\mu_p$$

$$\sigma = \sigma_n + \sigma_p$$

Where  $\sigma_n = nq\mu_n$  and  $\sigma_p = nq\mu_p$  the electrical conductivities due to electrons and holes respectively.

**Diffusion Currents:** A directed movement of charge carriers constitutes an electric current diffusion takes place due to the existence of a non-uniform concentration of carriers. Fick's

states that the diffusion flux  $F$  i.e. the particle current are proportional to and in a direction opposite to the gradient of the particles.

It can be written as

$$F \propto \frac{-\partial N}{\partial x}.$$

$$F = \frac{-D\partial N}{\partial x}$$

Where 'D' is diffusion current.

If 'n' and 'p' are the electron and hole concentrations, then the flux densities of electrons and holes  $J_e$  and  $J_p$  can be written as

$$J_e = -D_n \frac{\partial n}{\partial x}$$

$$J_p = J_h = -D_p \frac{\partial p}{\partial x}$$

Where  $D_n$  and  $D_p$  are diffusion constants of the electrons and holes respectively.

Then diffusion current densities become.

$$J_{ndiff} = qD_n \frac{\partial n}{\partial x}.$$

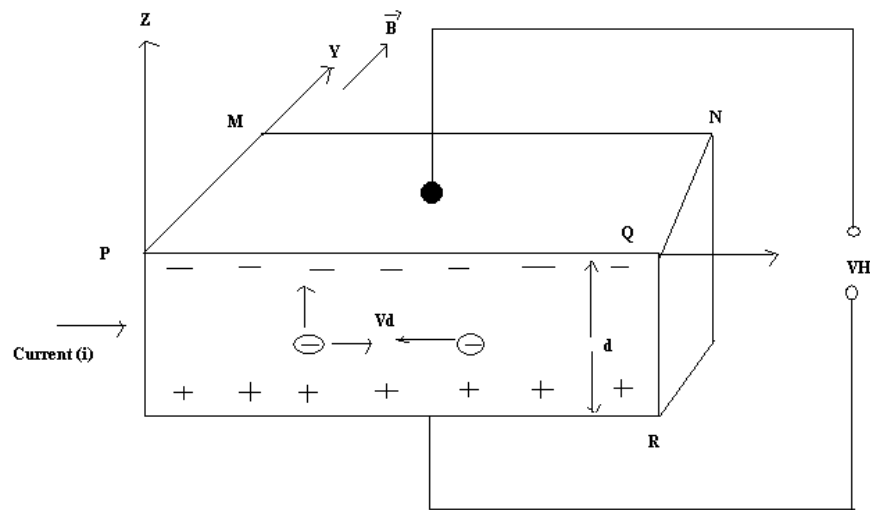
$$J_{pdiff} = -qD_p \frac{\partial p}{\partial x}$$

Where +q and -q are charges of the hole and electron respectively.

## 2.5 Hall Effect

When a magnetic field is applied perpendicular to a current carrying conductor or semiconductor, a voltage is developed across the specimen in a direction perpendicular to both the current and the magnetic field. This phenomenon is called the Hall Effect and the voltage developed is called the Hall voltage.

Consider a uniform thick metal strip placed with its length parallel to x-axis. Let a current 'i' is passed in the conductor along x-Axis and magnetic field B is established along y-axis. Due to the magnetic field charge carriers experience a force  $F_D$  perpendicular to x-y plane. i.e. along z-axis. This force direction is given by Fleming left hand rule.



The nature of charge carriers can be found by determining the sign of Hall e.m f which can be measured by potentiometers. Experiments showed that the charge carriers in metal are electrons while the charge carriers in p-type semiconductors are holes.

$$\text{Magnetic deflecting force } F_D = q(v_d \times B) \text{-----(1)}$$

$$\text{Hall electric deflecting force } F_H = qE_H \text{-----(2)}$$

When equilibrium is reached, the magnetic deflecting forces on the charge carriers are balanced by the electric forces due to electric field.

Hence net force on the charge carriers becomes zero.

From (1) & (2)

$$q(v_d \times B) + qE_H = 0$$

$$E_H = -(V_d \times B)$$

$$\text{In terms of magnitude } E_H = V_d \cdot B \text{-----(3)}$$

Where  $V_d$  is drift velocity of electrons.

The relation b/w current density and drift velocity is  $J = V_d n e$

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$$V_d = \frac{J}{ne} = \frac{J}{nq} \text{-----(4)}$$

Where 'n' is the number of charge carriers per unit volume

Substitute Eq'n (4) in equation (3)

$$E_H = \left( \frac{J}{nq} \right) B \text{-----(5)}$$

If  $V_H$ - be the Hall voltage in equilibrium.

$$\therefore \text{ Hall electric field, } E_H = \left( \frac{V_H}{d} \right) \text{-----(6)} \left( \because E = \frac{V}{d} \right)$$

Where 'd' is width of the metal strip.

By measuring the current in the slab, the current density J can be calculated.  $\left( J = \frac{i}{A} \right)$

Where 'A' is the area of cross section of the slab.

The magnetic field 'B' can be measured by Gauss-meter.

On substituting  $E_H$ , J and B in Equation (5) the value of  $\frac{1}{nq}$  can be calculated

$$\therefore \text{ Hall coefficient } R_H = \frac{E_H}{JB}$$

$$\frac{E_H}{JB} = \frac{1}{nq} \therefore R_H = \frac{1}{nq}$$

The Hall coefficient is negative when the charge carriers are electrons and positive when the charge carriers are holes.

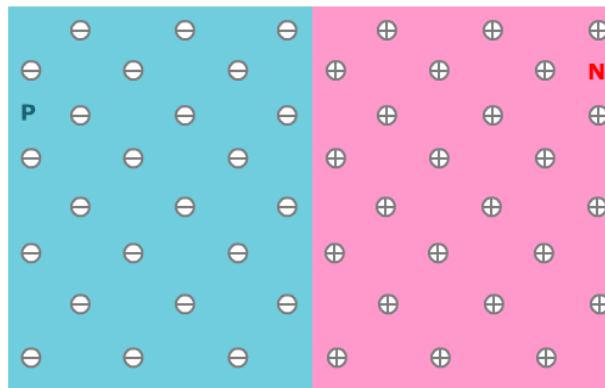
### Applications of Hall Effect:

1. The sign (electrons or holes) of charge carriers is determined.
2. The carrier concentration can be determined.
3. The mobility of charge carriers is measured directly.
4. Electrical conductivity of the material can be determined



5. It can be used to determine whether the given material is metal, insulator or semiconductor.
6. We can measure the unknown magnetic field by using the measured Hall voltage and the Hall coefficient for the slab.

## 2.6 PN JUNCTION DIODE



**Fig:** P- type material is joined with N-type material

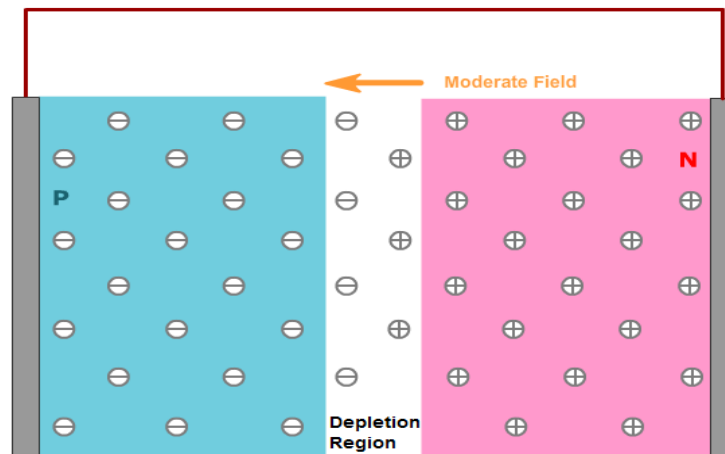
### Carrier Movement in Equilibrium:

A p-n junction with no external inputs represents equilibrium between carrier generation, recombination, diffusion and drift in the presence of the electric field in the depletion region. Despite the presence of the electric field, which creates an impediment to the diffusion of carriers across the electric field, some carriers still cross the junction by diffusion.

In the Figure below, most majority carriers which enter the **depletion region** move back towards the region from which they originated. However, statistically some carriers will have a high velocity and travel in a sufficient net direction such that they cross the junction. Once a majority carrier crosses the junction, it becomes a minority carrier. It will continue to diffuse away from the junction and can travel a distance on average equal to the diffusion length before it recombines.

The current caused by the diffusion of carriers across the junction is called diffusion current. In the figure below, watch the carriers in the depletion region and wait for carriers which cross the junction. Remember that in an actual p-n junction the number and velocity of the carriers is much greater and that the number of carriers crossing the junction is much larger.

Minority carriers which reach the edge of the diffusion region are swept across it by the electric field in the depletion region. This current is called the drift current. In equilibrium the drift current is limited by the number of minority carriers which are thermally generated within a diffusion length of the junction.



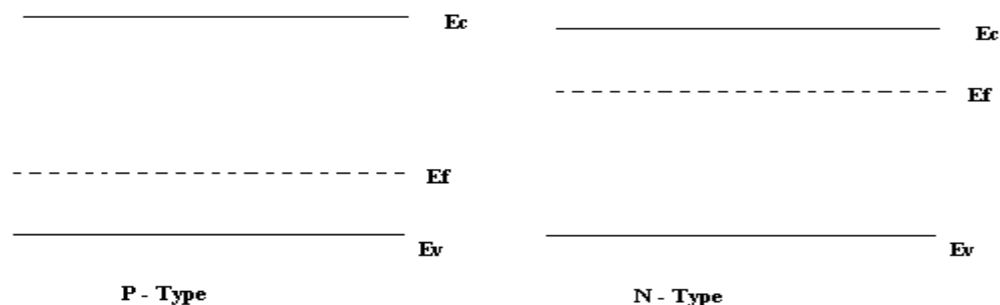
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**Fig:** Formation of PN junction

In equilibrium, the net current from the device is zero. The electron drift current and the electron diffusion current exactly balance out (if they did not there would be a net buildup of electrons on either one side or the other of the device). Similarly, the hole drift current and the hole diffusion current also balance each other out.

### Energy Diagrams of PN Diode:

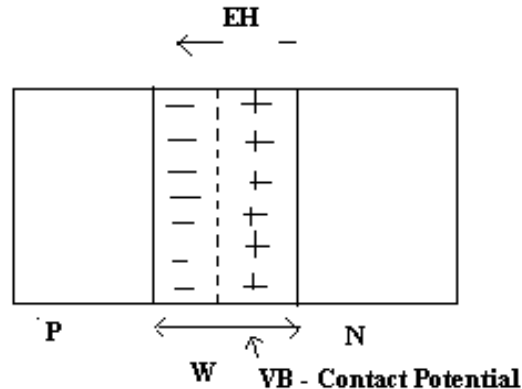
1. Before the formation of PN junction, the Fermi level lies to valence band in P type and near to conduction band in N type.



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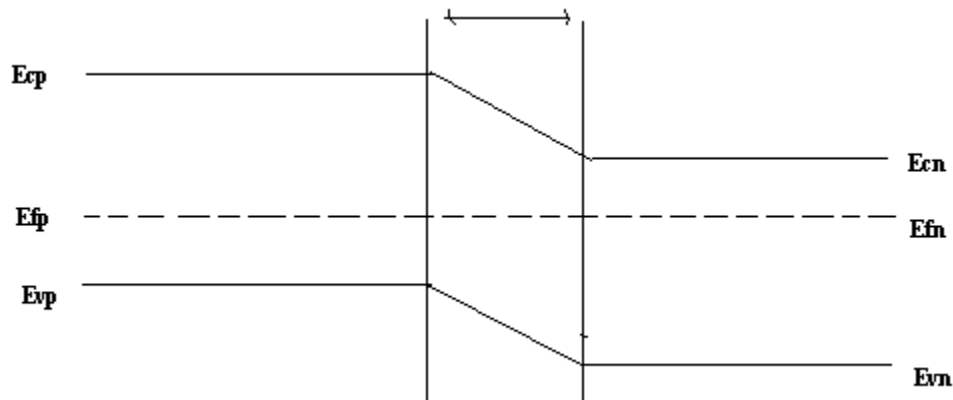
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2. When a P type material is brought closer to N type material, a contact potential  $V_B$  is developed which is due to potentials of depletion regions on n- side p- side respectively.

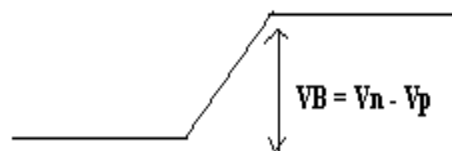


3. The contact or barrier potential due to n- type is  $V_n$  and p-type is  $V_p$ . There fore at the junction  $V_B = V_n - V_p$

4. After the junction formation, the Fermi level becomes common for both types.



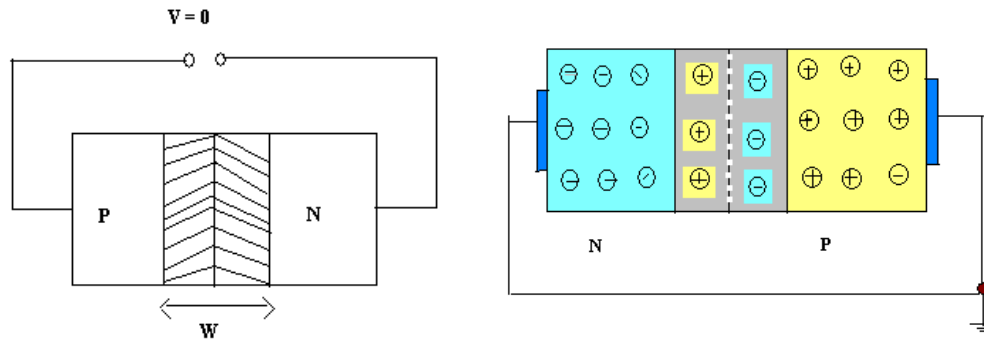
$$\begin{aligned} \text{The electric field } E_B &= E_{vp} - E_{vn} \\ &= E_{cp} - E_{cn} \\ &= eV_B \end{aligned}$$



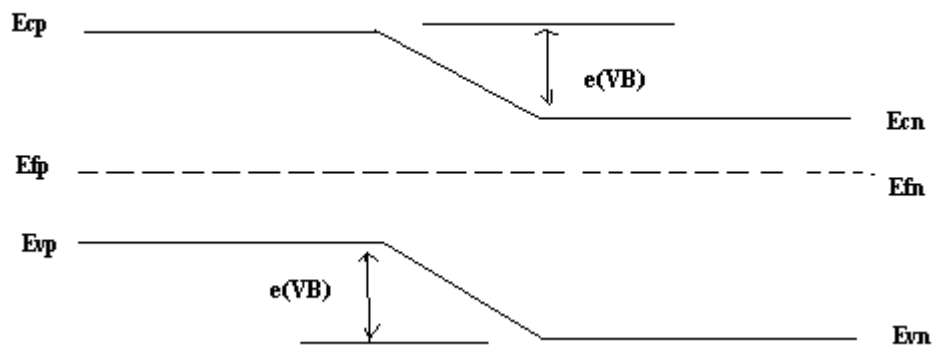
There are three biasing conditions for the standard junction diodes. They are,

1. Zero Bias
2. Forward Bias
3. Reverse Bias

**1. Zero Bias:** When a diode is said to be zero biased, no external voltage is applied to the PN Junction.

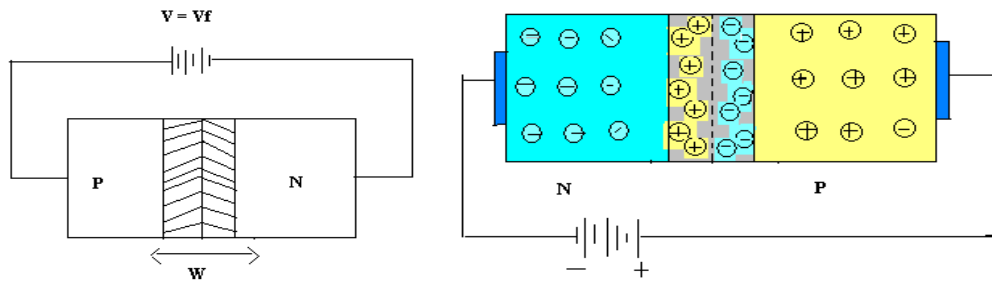


**Fig:** Zero biased PN junction

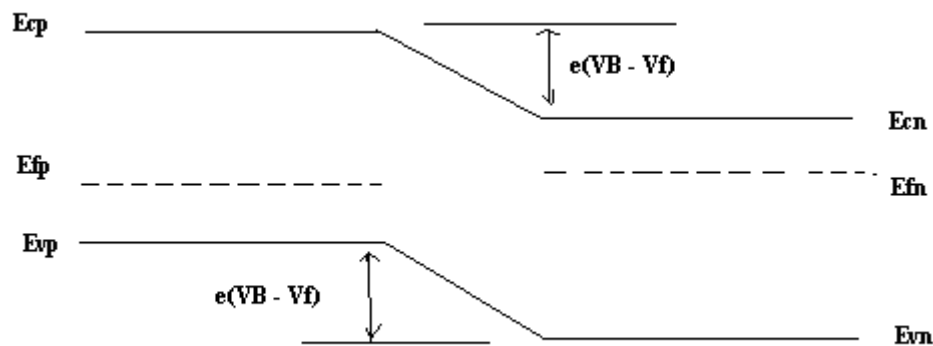


**Fig:** Energy level diagram

**2. Forward Bias:** When the positive terminal of the battery is connected to the P side and negative terminal to the N side, then the PN junction is known as Forward Biased.

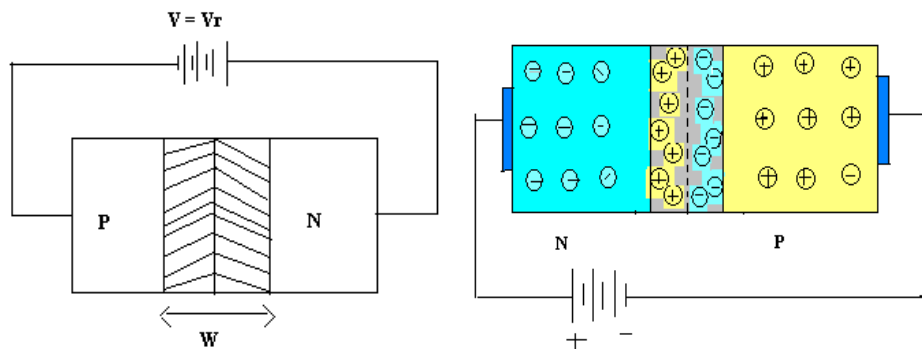


**Fig:** Forward biased PN junction

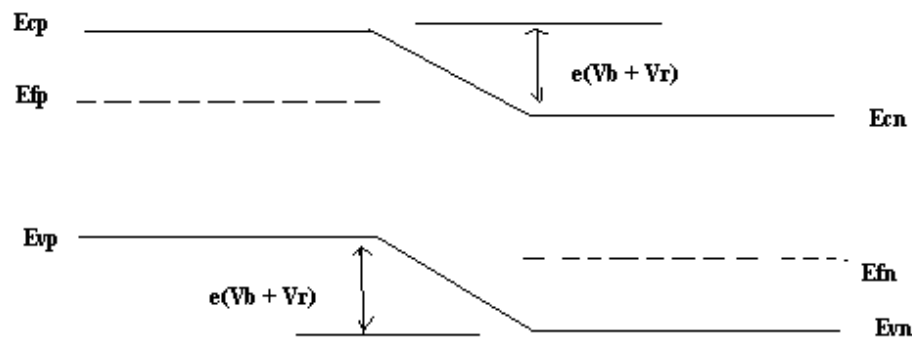


**Fig:** Energy level diagram

**3. Reverse Bias:** When the positive terminal of the battery is connected to the N side and negative terminal is to the P side then the PN junction is known as Reverse Biased.



**Fig:** Reverse biased PN junction



**Fig:** Energy level diagram

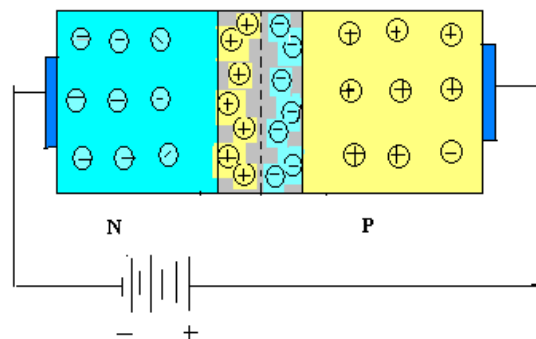
## I-V Characteristics of PN Junction:

To understand how a diode responds when it is connected in an electrical circuit, its  $V - I$  characteristics should be studied.

It can be understood simply by plotting a graph between the voltage applied across the terminals of the diode and the current that flows through the diode.

### Forward Characteristics:

1. When a diode is connected in a forward bias condition the current through the diode is small for the low voltage. i.e. for first few tenths of volts.

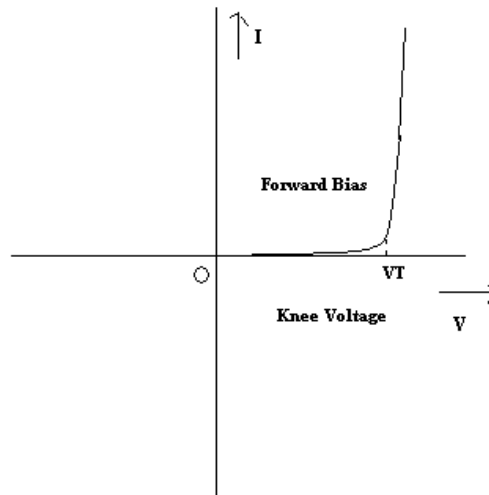


2. The external voltage applied reduces the barrier potential at the junction, but the diode does not conduct well until the applied external voltage overcomes the barrier potential.

3. When this happens, large number of electrons from n- side and holes from p-side start crossing the junction.

4. Now, even small increase in applied voltage produces a sharp increase in current.

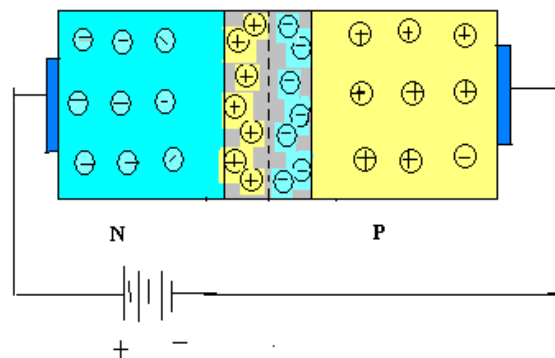
5. The voltage beyond which the current increases rapidly is called **Knee voltage or Threshold voltage**.



6. Since the diode can conduct infinite current above this knee point as if effectively becomes a short circuit, resistors are used in series with the device limit its current flow.

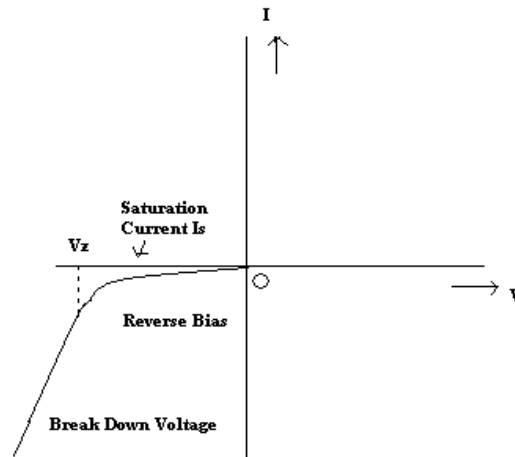
### Reverse Characteristics:

1. Under Reverse bias condition, the conduction through the diode is almost negligible.

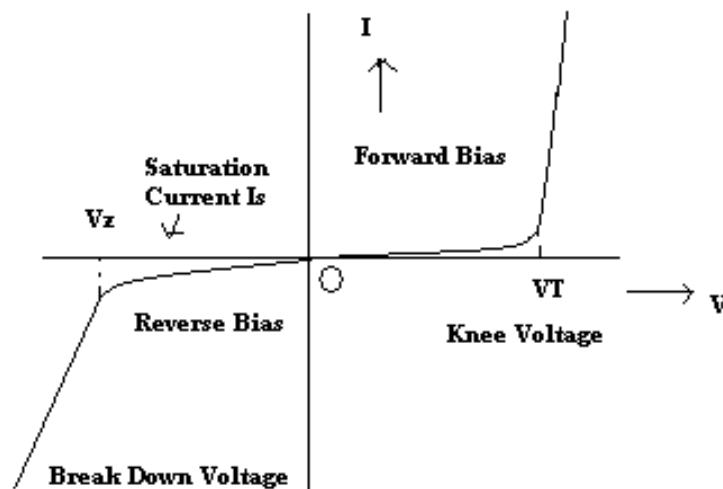


2. A very small current flows through the diode due to minority carriers crossing the junction.

3. This current remains constant for all voltages less than the break down voltage  $V_z$ . This current is called **reverse saturation current**.



4. When the applied reverse bias voltage becomes more than the break down voltage  $V_z$ , the current increases at a rapid rate.
5. An ordinary PN junction should not be operated in this voltage, because the diode becomes overheat and failed due to the “avalanche effect” around the junction.



**Fig:** I – V Characteristics of PN junction diode

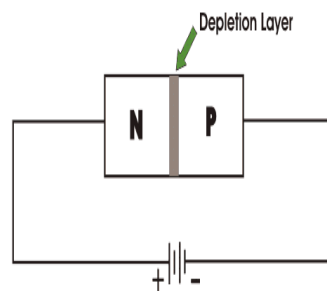


## 2.7 Zener diode:

A zener diode is a special type of device designed to operate in the zener breakdown region. Zener diodes act like normal p-n junction diodes under forward biased condition. The breakdown voltage of a zener diode is carefully set by controlling the doping level during manufacture. Zener diodes are mainly used to protect electronic circuits from over voltage.

Zener diode is heavily doped than that of the normal p-n junction diode. Hence, it has very thin depletion region. Therefore, zener diodes allow more electric current than the normal p-n junction diodes. Zener diode allows electric current in forward direction like a normal diode but also allows electric current in the reverse direction if the applied reverse voltage is greater than the zener voltage. Zener diode is always connected in reverse direction because it is specifically designed to work in reverse direction.

The name zener diode was named after the American physicist Clarence Melvin Zener who discovered the zener effect. Zener diodes are the basic building blocks of electronic circuits. They are widely used in all kinds of electronic equipments.



### Breakdown in zener diode

There are two types of reverse breakdown regions in a zener diode: avalanche breakdown and zener breakdown.

**Avalanche breakdown:** The avalanche breakdown occurs in both normal diodes and zener diodes at high reverse voltage. When high reverse voltage is applied to the p-n junction diode, the free electrons (minority carriers) gain large amount of energy and accelerated to greater velocities.

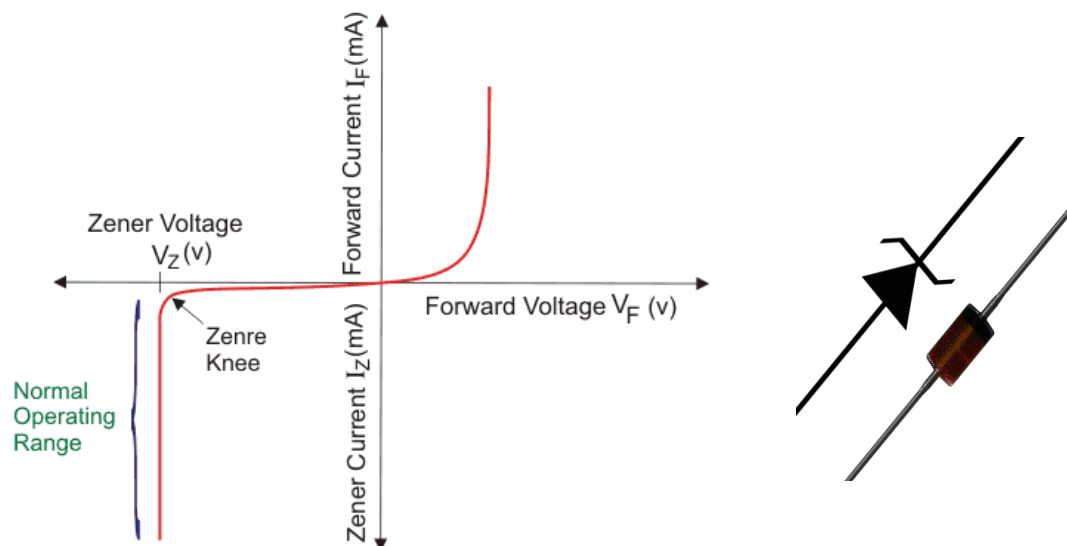
The free electrons moving at high speed will collide with the atoms and knock off more electrons. These electrons are again accelerated and collide with other atoms. Because of this continuous collision with the atoms, a large number of free electrons are generated. As a result, electric current in the diode increases rapidly. This sudden increase in electric current may permanently destroy the normal diode. However, avalanche diodes may not be destroyed because they are carefully designed to operate in avalanche breakdown region. Avalanche breakdown occurs in zener diodes with zener voltage ( $V_z$ ) greater than 6V.

**Zener breakdown:** The zener breakdown occurs in heavily doped p-n junction diodes because of their narrow depletion region. When reverse biased voltage applied to the diode is increased, the narrow depletion region generates.

When reverse biased voltage applied to the diode reaches close to zener voltage, the electric field in the depletion region is strong enough to pull electrons from their valance band. The valence electrons which gains sufficient energy from the strong electric field of depletion region will breaks bonding with the parent atom will become free electrons. This free electrons carry electric current from one place to another place. At zener breakdown region, a small increase in voltage will rapidly increases the electric current.

- Zener breakdown occurs at low reverse voltage whereas avalanche breakdown occurs at high reverse voltage.
- Zener breakdown occurs in zener diodes because they have very thin depletion region.
- Breakdown region is the normal operating region for a zener diode.
- Zener breakdown occurs in zener voltage ( $V_Z$ ) less than 6V. Now, discussing about the diode circuits we should look through the graphical representation of the operation of the **zener diode**. Normally, it is called the V-I characteristics of a Zener diode.

#### V-I Characteristics of zener diode



The VI characteristics of a zener diode are shown in the above figure and its symbol. When forward biased voltage is applied to the zener diode, it works like a normal diode. However, when reverse biased voltage is applied to the zener diode, it works in different manner. When reverse biased voltage is applied to a zener diode, it allows only a small amount of leakage current until the voltage is less than zener voltage. When reverse biased voltage applied to the zener diode reaches zener voltage, it starts allowing large amount of electric current. At this

point, a small increase in reverse voltage will rapidly increases the electric current. Because of this sudden rise in electric current, breakdown occurs called zener breakdown. However, zener diode exhibits a controlled breakdown that does damage the device. The zener breakdown voltage of the zener diode is depends on the amount of doping applied. If the diode is heavily doped, zener breakdown occurs low reverse voltages.

On the other hand, if the diode is lightly doped, the zener breakdown occurs at high reverse voltages.

Zener diodes are available with zener voltages in the range of 1.8V to 400V.

### **Advantages of zener diode:**

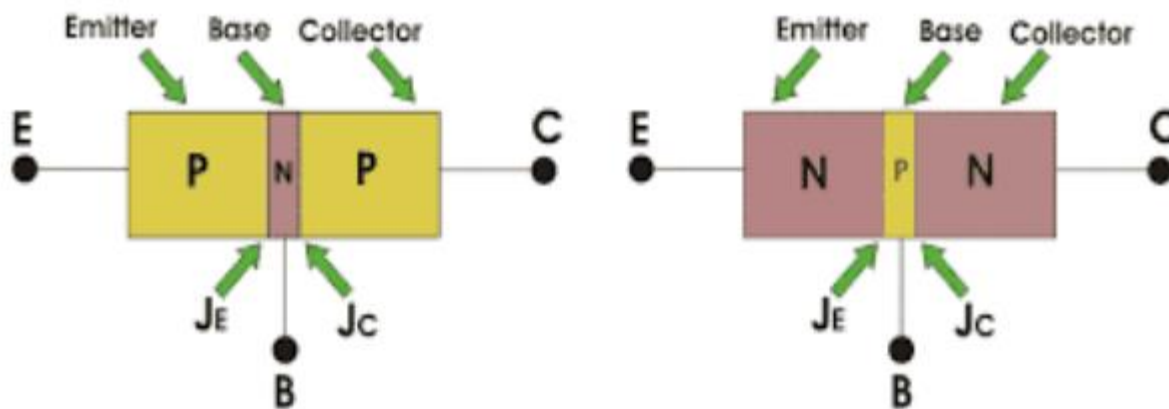
- Power dissipation capacity is very high
- High accuracy
- Small size
- Low cost

### **Applications of zener diode:**

- It is normally used as voltage reference
- Zener diodes are used in voltage stabilizers or shunt regulators.
- Zener diodes are used in switching operations
- Zener diodes are used in clipping and clamping circuits.
- Zener diodes are used in various protection circuits.

## **2.8 Bipolar Junction Transistor (BJT) Construction:**

A Bipolar Junction Transistor (BJT) has three terminals connected to three doped semiconductor regions. In an NPN transistor, a thin and lightly doped P-type base is sandwiched between a heavily doped N-type emitter and another N-type collector while in a PNP transistor, a thin and lightly doped N-type base is sandwiched between a heavily doped P-type emitter and another P-type collector. In the following we will only consider NPN BJTs.



**1. Emitter:** Emitter terminal is the heavily doped region as compared two base and collector. This is because the work of the emitter is to supply charge carrier to the collector via the base. The size of the emitter is more than base but less than the collector.

**2. Base:** The size of base region is extremely small, it is less than emitter as well as the collector. The size of the base is always kept small so that charge carriers coming from the emitter and entering base will not recombine in the base region and will be directed towards the collector region. The doping intensity of base is also less than emitter and collector for the same reason mentioned above.

**3. Collector:** The collector terminal is moderately doped, and the size of the collector region is slightly more than emitter region because all the charge carriers coming from the emitter recombine at base and heat is released in this process. Thus, it is necessary for the collector terminal to be large enough so that it can be dissipate the heat and the device may not burn out.

### Principle of operation:

In many schematics of transistor circuits (especially when there exist a large number of transistors in the circuit), the circle in the symbol of a transistor is omitted. The figures below show the cross section of two NPN transistors. Note that although both the collector and emitter of a transistor are made of N-type semiconductor material, they have totally different geometry and therefore cannot be interchanged. All previously considered components (resistor, capacitor, inductor, and diode have two terminals (leads) and can therefore be characterized by the single relationship between the current going through and the voltage across the two leads. Differently, a transistor is a three-terminal component, which could be considered as a two-port network with an input-port and an output-port, each formed by two of the three terminals, and characterized by the relationships of both input and output currents and voltages. Depending on which of the three terminals is used as common terminal, there can be three possible

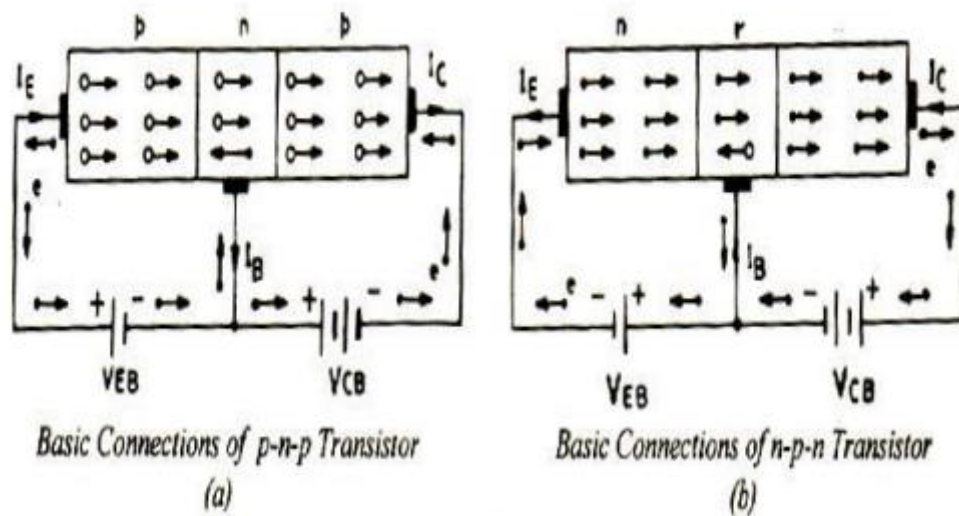
### Configuration for the two-part network formed by a transistor:

- Common emitter (CE)

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- Common base (CB)
- Common Collector (CC)



- Common-Base (CB) configuration

Two voltages  $V_{BE}$  and  $V_{CB}$  are applied respectively to the emitter E and collector C, with respect to the common base B, so that the BE junction forward biased while the CB junction is reversed biased. Note that the polarity of  $V_{BE}$  and direction of  $I_B$  associated with the PN-junction between E and B are the same as those associated with a diode, voltage polarity: positive on P, negative on N, current direction: from P to N, but  $V_{CB}$  and the direction of  $I_C$  associated with the PN-junction between the base and collector are defined oppositely.

### The behaviour of the NPN-transistor is determined by its two PN-junctions:

- The forward biased base-emitter (BE) PN-junction allows the free electrons in emitter to go through the PN-junction to arrive at the base, forming the emitter current  $I_E$ .
- As the P-type base is thin and lightly doped, only a small number of the electrons from the emitter are combined with the holes in base to form the base current  $I_B$ .
- Most of the electrons coming from the emitter become minority carrier in the P-type base, and they go through the reverse biased collector-base PN junction to arrive at the collector.
- The percentage of those electrons that arrive at the collector out of the electrons from the emitter is defined as  $\alpha$  (e.g. 99%, depending on the doping and geometry of the material). The total collector current  $I_C$  is therefore  $I_C = \alpha I_E$ . The current gain or current transfer ratio is defined

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as the ratio between the emitter (input) current  $I_E$  and the collector (output)  $I_C$ .

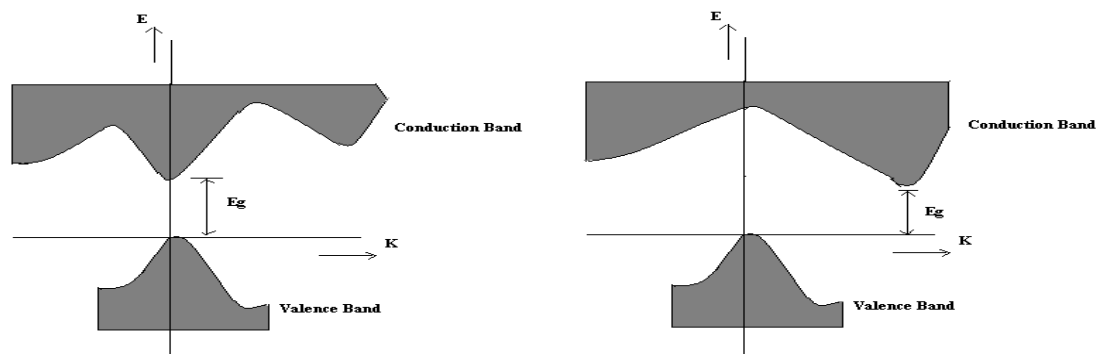
### Formulas:

|           |                                                                                                                                                         |          |                                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>01</b> | Fermi level in P-type semiconductor is<br>$E_f = \frac{E_v + E_a}{2}$                                                                                   | <b>5</b> | Density of electrons in the conduction band is<br>$n = (2N_d)^{1/2} \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/4} \exp \left( \frac{E_d - E_c}{2kT} \right)$ |
| <b>02</b> | The diode equation is given by<br>$I = I_0 \left[ \exp \left( \frac{eV}{\beta kT} \right) - 1 \right]$                                                  | <b>6</b> | Fermi level in N-type semiconductor is<br>$E_f = \frac{E_d + E_c}{2}$                                                                                         |
| <b>03</b> | Density of electrons in a semiconductors is given by<br>$n = 2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} \exp \left( \frac{E_f - E_c}{kT} \right)$ | <b>7</b> | <b>Fermi level in intrinsic semiconductor is</b><br>$E_f = E_g/2$                                                                                             |
| <b>04</b> | Density of holes in a semiconductor is given by<br>$p = 2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} \exp \left( \frac{E_v - E_f}{kT} \right)$      | <b>8</b> | Intrinsic carrier concentration is given by<br>$n_i = 2 \left( \frac{2\pi kT}{h^2} \right)^{3/2} (m_e^* m_h^*)^{3/4} \exp \left( -\frac{E_g}{2kT} \right)$    |

## SEMICONDUCTOR DEVICES

### 2.9 Direct and Indirect Band Gap Semiconductors:

Semiconductors are available in elemental form (Ge,Si) and compound form (GaAs,Inp) Further they can further subdivided into direct band gap and indirect band gap semiconductors.



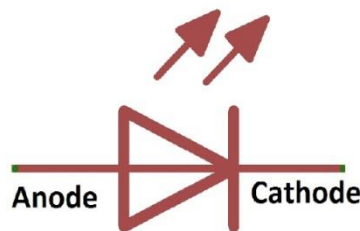
**Fig:** (a) Band diagram of direct band gap and indirect band gap semiconductors

| Direct band gap Semiconductor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Indirect band gap Semiconductor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>Fig(a) shows the energy band diagram drawn between energy 'E' and wave vector 'k' of a direct band gap semi conductor.</li> <li>As shown in the band diagram, the minimum energy of conduction band and maximum energy of valence band are having the same value of wave vector.</li> <li>An electron from the conduction band can recombine with a hole the valence band directly emitting a light photon of energy <math>h\nu</math>.</li> <li>Life time (i.e. recombination time) of charge carriers is very less.</li> <li>Due to emission of light photon during recombination of charge carriers, these are used to fabricate LEDS and laser diodes.</li> <li>These are mostly from the</li> </ol> | <ol style="list-style-type: none"> <li>Fig (b) shows the energy band diagram drawn between 'E' and wave vector 'k' of an indirect band gap semiconductor</li> <li>In this band diagram the minimum energy of conduction band and maximum energy of valence band are having the different values of wave vector.</li> <li>An electron from the conduction band can recombine with a hole in valence band indirectly through traps. Here there is no emission of light photon.</li> <li>Life time of charge carriers is more</li> <li>Due to longer life time of charge Carriers, these are used to amplify the signals, as in the case of diodes and transistors.</li> <li>These are mostly from the elemental semiconductors</li> </ol> |

|                                                                                                                           |                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <p>compound semiconductors.</p> <p>7. Ex: - In P, GaAs.</p> <p>8. Band gap energy of InP =1.35eV<br/>And GaAs =1.42eV</p> | <p>7. Ex: - Si, Ge.</p> <p>8. Band gap energy of Si=0.7eV and<br/>Ge=1.12eV.</p> |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|

## 2.10 LIGHT EMITTING DIODE(LED):

The LED is a PN-junction diode which emits light when an electric current passes through it in the forward direction. In the LED, the recombination of charge carrier takes place. The electron from the N-side and the hole from the P-side are combined and gives the energy in the form of heat and light.



**Construction of LED:** The recombination of the charge carrier occurs in the P-type material, and hence P-material is the surface of the LED. For the maximum emission of light, the anode is deposited at the edge of the P-type material. The cathode is made of gold film, and it is usually placed at the bottom of the N-region. This gold layer of cathode helps in reflecting the light to the surface.

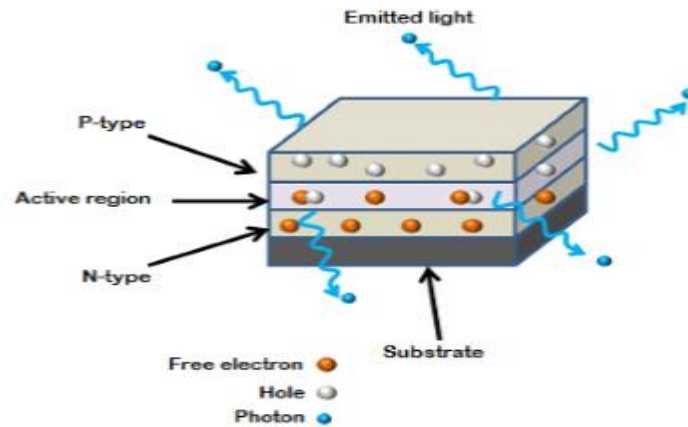
The gallium arsenide phosphide is used for the manufacturing of LED which emits red or yellow light for emission. The LED are also available in green, yellow amber and red in colour.

### Working of LED:

The working of the LED depends on the quantum theory. The quantum theory states that when the energy of electrons decreases from the higher level to lower level, it emits energy in the form of photons. The energy of the photons is equal to the gap between the higher and lower level.

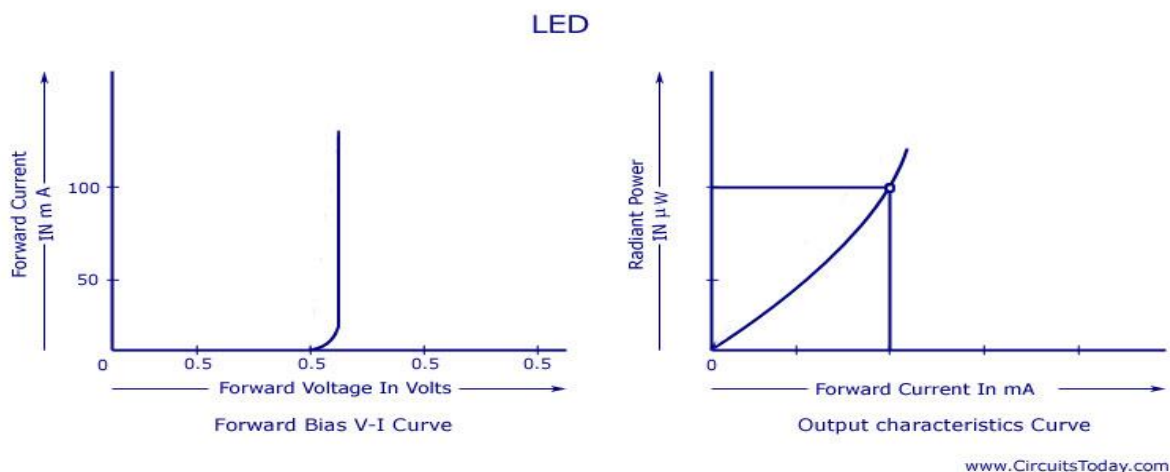
The LED is connected in the forward biased, which allows the current to flows in the forward direction. The flow of current is because of the movement of electrons in the opposite direction. The recombination shows that the electrons move from the conduction band to valence band and they emits electromagnetic energy in the form of photons. The energy of photons is equal to the gap between the valence and the conduction band.





The gallium arsenide phosphide is used for the manufacturing of LED which emits red or yellow light for emission. The LED are also available in green, yellow amber and red in colour.

V –I Characteristics of LED:



The output characteristics curve shows that radiant power of LED is directly proportional the forward current in LED.

Applications of LEDs

- Electronic displays such as OLEDs, micro-LEDs, quantum dots etc.
- As an LED indicator.
- In remote controls.
- Lightings.
- Opto-isolators.

## 2.12 SEMICONDUCTOR LASER

A **semiconductor diode laser** is a specifically made p-n junction diode that emits coherent light under forward bias. R.N. Hall and his coworkers made the first semiconductor laser in 1962. P-N junction lasers emit light almost anywhere in the spectrum from UV to IR.

The **laser diode** (light amplification by stimulated emission of radiation) produces a monochromatic (single Color) light. Laser diodes in conjunction with photodiodes are used to retrieve data from compact discs.

Diode lasers are remarkably small in size (0.1 mm long). They have high efficiency of the order of 40%. In spite of their small size and low power requirements, they produce power outputs equivalent to that of He-Ne lasers. Diode lasers are useful in optical fibre communications, in CD players, CD-ROM drives, optical reading and high speed laser printing etc wide.

### a) Semiconductor Materials

- i) Semiconductors are two different groups, direct band gap semiconductors and indirect band gap semiconductors.
- ii) Direct band gap semiconductors are formed by group III\_V elements and group IV-VI elements. Most of the compound semiconductors belong to this group.
- iii) Lasers are made using direct band gap semiconductors. Gallium Arsenide (GaAs) diode is an example of semiconductor diode laser.
- iv) Direct band gap semiconductor is the one in which a conduction band electron can recombine directly with a hole in the valence band. The recombination process leads to emission of light.

### b) Principle

- i) The energy band structure of a semiconductor consists of a valence band and a conduction band separated by an energy gap,  $E_g$ . The conduction band contains electrons and the valence band contains holes and electrons.
- ii) When an electron from the conduction band jumps into a hole in the valence band, the excess energy  $E_g$  is given out in the form of a photon.
- iii) Electron-hole recombination is the basic mechanism responsible for emission of light.
- iv) The wavelength of the light is given by the relation  $\lambda = hc/E_g$ .
- v) Semiconductors having a suitable value of  $E_g$  emit light in the optical region.

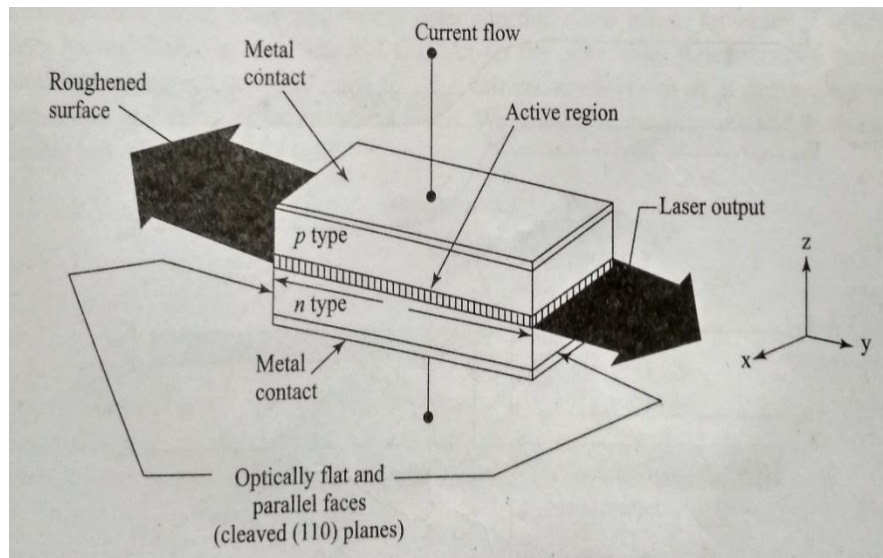
### c) Types of Semiconductor Diode Lasers

Broadly there are two types of semiconductor diode lasers. They are known as *homojunction semiconductor laser* and *heterojunction semiconductor lasers*.

- i) Homojunction Semiconductor Laser
  - ii) Heterojunction Semiconductor Laser
- Homojunction Semiconductor Laser**

A Simple diode laser which makes use of the same semiconductor material on both sides of the junction is known as a homojunction diode laser. Example: Gallium arsenide (GaAs) laser.

**a) Construction:** Fig. shows the schematic of a homojunction diode laser. Starting with a heavily doped n-type GaAs material, a p-region is formed on its top by diffusing zinc atoms into it. A heavily zinc doped layer constitutes the heavily doped p-region.



**Fig:** Homojunction diode laser

The diode is extremely small in size. Typical diode chips are  $500\text{ }\mu\text{m}$  long and about  $100\text{ }\mu\text{m}$  wide and thick. The top and bottom faces are metalized and metal contacts are provided to pass current through the diode. A pair of parallel planes cleaved at the two ends of the PN junction provides required reflection to form the cavity. The two remaining sides of the diode are roughened to remove lasing action in that direction. The entire structure is packaged in small case which looks like the metal case used for discrete transistors.

### **b) Working:**

- i) Heavily doped p- and n- regions are used. Because of very high doping on n-side, the donor levels are broadened and extend into the conduction band.
- ii) The Fermi level is pushed into the conduction band and electrons occupy the levels lying below the Fermi level.
- iii) Similarly, on the heavily doped p-side the Fermi level lies within the valence band and holes occupy the portion of the valence band that lies above the Fermi Level.
- iv) At thermal equilibrium, the Fermi level is uniform across the junction.

## 2.13 Photo Detectors

A photo detector is an optoelectronic device which absorbs light and converts the optical energy into measurable electric current.

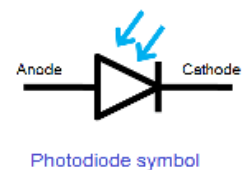
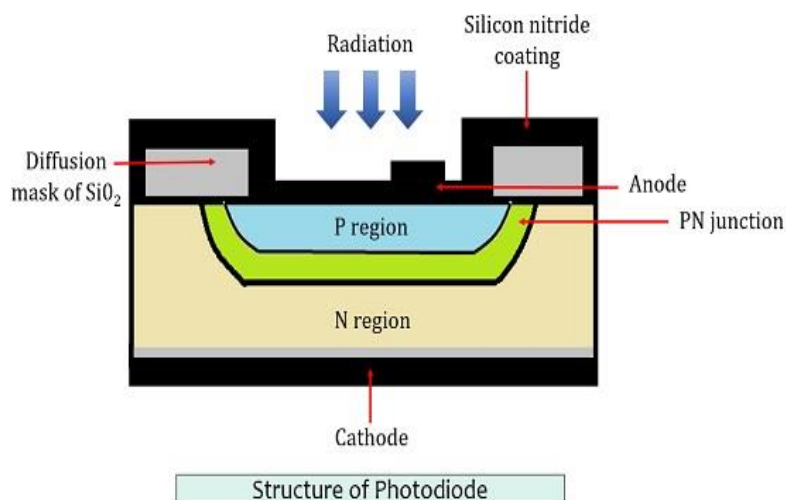
**Photodiode:** A photodiode is a p-n junction diode, used to convert the light into current or voltage by operating in reverse biased mode.

### Working Principle:

The working principle of a photodiode is, when a photon of ample energy strikes the diode, it makes a couple of an electron-hole pairs. Therefore, holes in the region move toward the anode, and electrons move toward the cathode, and a photocurrent will be generated.

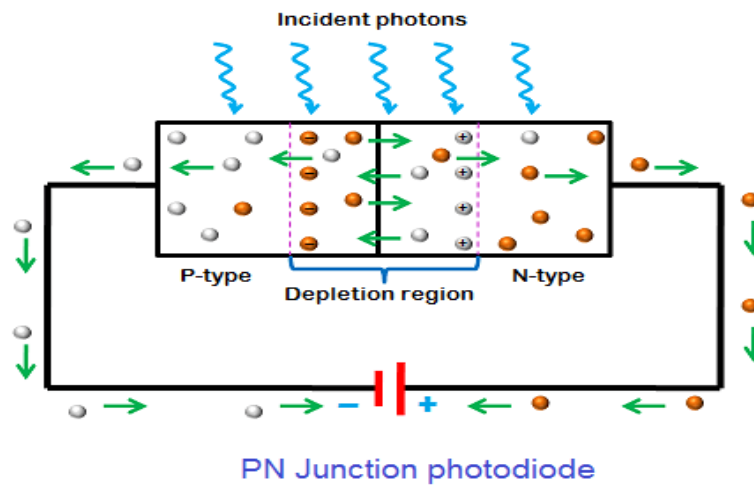
**Construction:** The photodiode is made up of two layers of P-type and N-type semiconductor. In this, the P-type material is formed from diffusion of the lightly doped P-type substrate. Thus, the layer of P<sup>+</sup> ions is formed due to the diffusion process. And N-type epitaxial layer is grown on N-type substrate. The contacts are made up of metals to form two terminal cathode and anode. The front area of the diode is divided into two types that are active surface and non-active surface. The non-active surface is made up of SiO<sub>2</sub> (Silicon di Oxide) and the active surface is coated with anti-reflection material. The active surface is called so because the light rays are incident on it.

While on the non-active surface the light rays do not strike. The active layer is coated with anti-reflection material so that the light energy is not lost and the maximum of it can be converted into current.



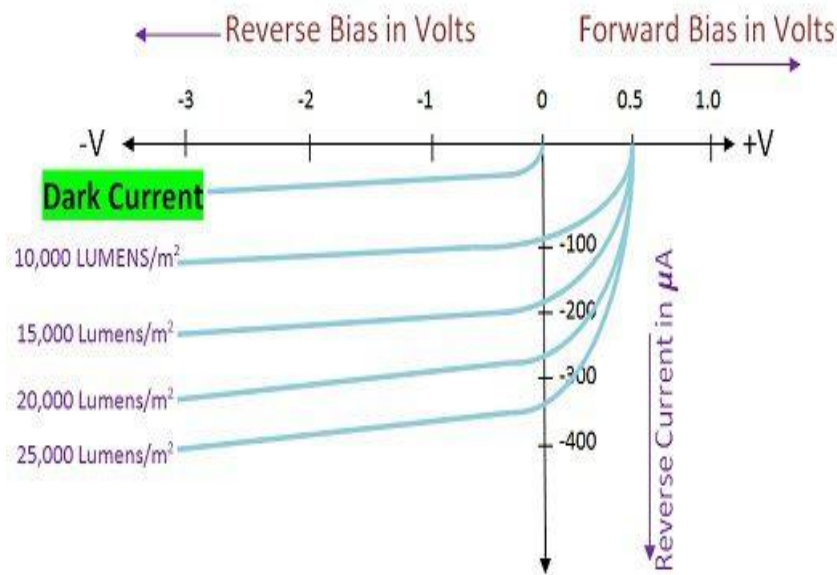
**Working:** When a light is made to illuminate the PN junction, the photon enters the depletion region of diode and hits the atom with high energy. This results electron - hole pairs. In general, an electron will have negative charge and holes will have a positive charge. The depletion energy

will have built in electric field. Due to that electric field, electron hole pairs moves away from the junction. Hence, holes move to anode and electrons move to cathode to produce photo current.



### V-I Characteristics of Photodiode

Photodiode operates in reverse bias condition. Reverse voltages are plotted along X axis in volts and reverse current are plotted along Y-axis in microampere. Reverse current does not depend on reverse voltage. When there is no light illumination, reverse current will be almost zero. The minimum amount of current present is called as **Dark Current**. Once when the light illumination increases, reverse current also increases linearly as shown in below figure.



### Applications of photodiodes

Photodiodes are used in

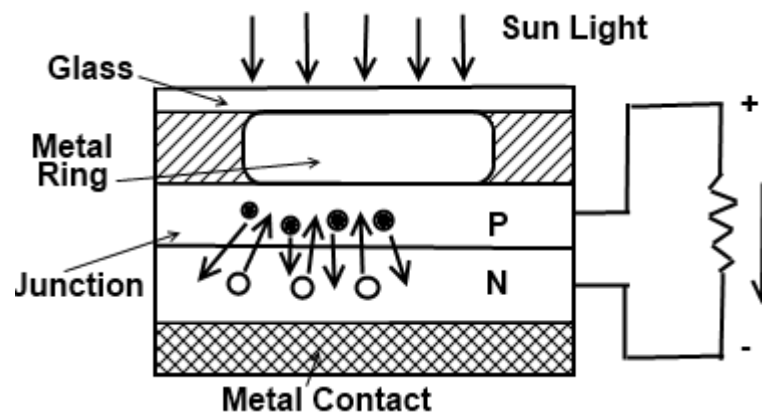
- Consumer electronics devices such as compact disc players,
- Smoke detectors,
- Medical devices
- Receivers for infrared remote control devices used to control equipment from televisions to air conditioners.

### 2.14 Solar Cell or Photovoltaic

A **solar cell** (also known as a photovoltaic cell or PV cell) is defined as an electrical device that converts light energy into electrical energy through the photovoltaic effect.

**Construction:** Consider the figure below shows the constructions of the silicon photovoltaic cell. The upper surface of the cell is made of the thin layer of the p-type material so that the light can easily enter into the material. The metal rings are placed around p-type and n-type material which acts as their positive and negative output terminals respectively.

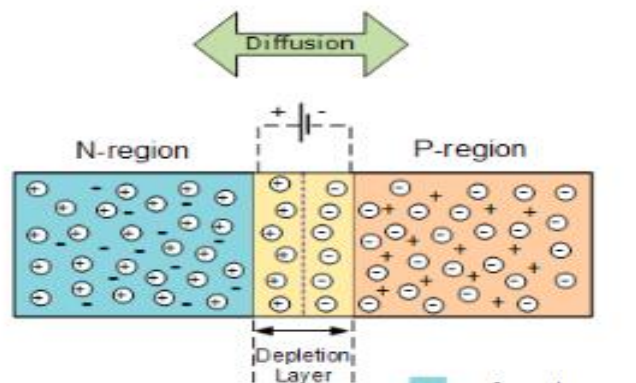
**Working Principle:** When light reaches the p-n junction, the light photons can easily enter in the junction, through very thin p-type layer. The light energy, in the form of photons, supplies sufficient energy to the junction to create a number of electron-hole pairs. The free electrons in the depletion region can quickly come to the n-type side of the junction. Similarly, the holes in the depletion can quickly come to the p-type side of the junction.



Once, the newly created free electrons come to the n-type side, cannot further cross the junction because of barrier potential of the junction. Similarly, the newly created holes once come to the p-type side cannot further cross the junction because of same barrier potential of the junction.

As the concentration of electrons becomes higher in one side, i.e. n-type side of the junction and concentration of holes becomes more in another side, i.e. the p-type side of the junction, the p-n junction will behave like a small battery cell. A voltage is set up which is known as photo

voltage. If we connect a small load across the junction, there will be a tiny current flowing through it.



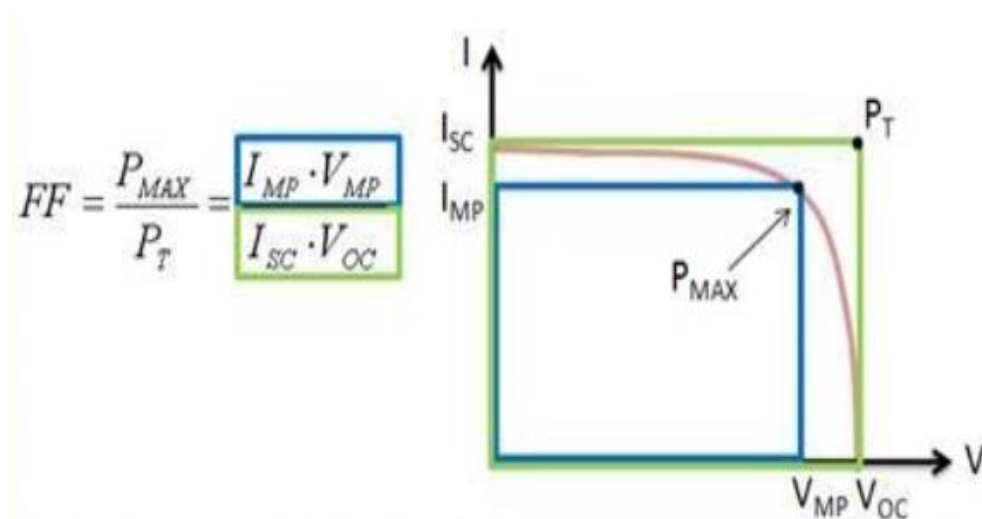
### V-I Characteristics of a solar Cell:

The voltage - current characteristics of solar cell is shown in below figure.

From the graph it is clear that the maximum current occurs in the solar cell when the voltage is zero is known as **short circuit current**  $I_{sc}$ .

The maximum voltage that occurs in the solar cell when the current is zero is known as **open circuit voltage**  $V_{oc}$ .

The **fill factor** is the ratio of the actual maximum obtainable power to the product of the open circuit voltage and short circuit current.



The efficiency of a solar cell is determined as the fraction of incident power which is converted to electricity and is defined as:



$$P_{max} = V_{OC} I_{SC} FF$$
$$\eta = \frac{V_{OC} I_{SC} FF}{P_{in}}$$

### Materials Used in Solar Cell:

The materials which are used for this purpose must have band gap close to 1.5ev. Commonly used materials are-

1. Silicon.
2. GaAs.
3. CdTe.
4. CuInSe<sub>2</sub>

### Criteria for Materials to be Used in Solar Cell

1. Must have band gap from 1ev to 1.8ev.
2. It must have high optical absorption.
3. It must have high electrical conductivity.
4. The raw material must be available in abundance and the cost of the material must be low.

### Advantages of Solar Cell

1. No pollution associated with it.
2. It must last for a long time.
3. No maintenance cost.

### Disadvantages of Solar Cell

1. It has high cost of installation.
2. It has low efficiency.
3. During cloudy day, the energy cannot be produced and also at night we will not get solar energy.



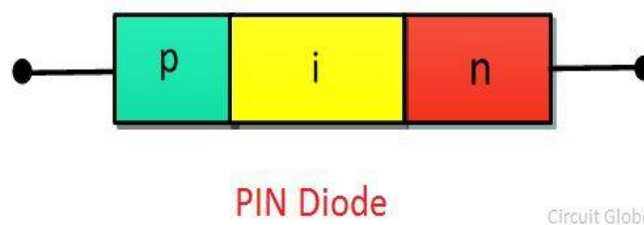
**Applications**

1. It may be used to charge batteries.
2. Used in light meters.
3. It is used to power calculators and wrist watches.
4. It can be used in spacecraft to provide electrical energy.

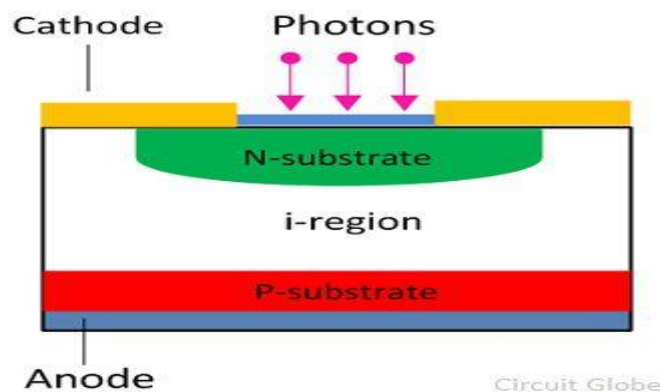
**2.15 PIN Diode**

The diode in which the intrinsic layer of high resistivity is sandwiched between the P and N-region of semiconductor material such type of diode is known as the PIN diode. The PIN diode is a type of photodetector used for converting the light energy into the electrical energy.

**Construction:** The diode consists the P-region and N-region which is separated by the intrinsic semiconductor material. The intrinsic region has no free charge carrier. It acts as an insulator between n and the p-type region. The i-region has the high resistance which obstructs the flow of electrons to pass through it.



**Working:** The working of the PIN diode is similar to the ordinary diode. When the diode is unbiased, their charge carrier will diffuse. The process of diffusion occurs continue until the charges become equilibrium in the depletion region.



Let the N and I-layer make the depletion region. The diffusion of the hole and electron across the region generates the depletion layer across the NI-region. The thin depletion layer induces across n-region, and thick depletion region of opposite polarity induces across the I-region.

When the diode is kept forward biased, the charges are continuously injected into the I-region from the P and N-region. This reduces the forward resistance of the diode, and it behaves like a variable resistance.

When the reverse voltage is applied across the diode, the width of the depletion region increases. The thickness of the region increases until the entire mobile charge carrier of the I-region swept away from it. The reverse voltage requires for removing the complete charge carrier from the I-region is known as the **swept voltage**. In reverse bias, the diode behaves like a capacitor. The P and N region acts as the positive and negative plates of the capacitor, and the intrinsic region is the insulator between the plates.

### Characteristics of PIN diode

- **Low Capacitance:** As we already discussed that a PIN diode offers a lower value of capacitance due to the larger distance between p and n region.
- **High breakdown voltage:** Due to the presence of intrinsic region, PIN diode exhibits a higher value of breakdown voltage.
- **High Sensitive to photodetection:** The depletion region is responsible for generating energy when radiation falls on its surface. The existence of an intrinsic region enhances the area for radiation absorption.
- **High Storage of carriers:** This is the important characteristic of PIN diode. The intrinsic region enhances the area for storage of carriers. The stored charge in the depletion region is responsible for the amount of current flowing through the circuit.

### Applications of PIN diode

1. **As a Photodetector:** A PIN diode can change the absorbed light into electrical energy. The placement of intrinsic region between the p and n region increases the region for radiation absorption. With the increased radiation absorption region, the efficiency of the device to produce electrical energy also increases. Thus, it can be used as a photodiode.
2. **As a radio frequency switch:** The intrinsic region isolates the p and n region of the diode due to which capacitance decreases. The capacitance of the device should almost negligible in order to operate it as a switch.
3. **As a voltage rectifier:** PIN diode is able to bear high reverse voltage due to the intrinsic layer. This leads to an increase in the breakdown voltage of the diode. Hence, due to this, the device allows the rectification of high input voltage.

## 2.16 Avalanche Diode

An avalanche diode is a special type of semiconductor device designed to operate in reverse breakdown region.

**Construction:** Avalanche diodes are generally made from silicon or other semiconductor materials. The construction of avalanche diode is similar to zener diode but the doping level in avalanche diode differs from zener diode.

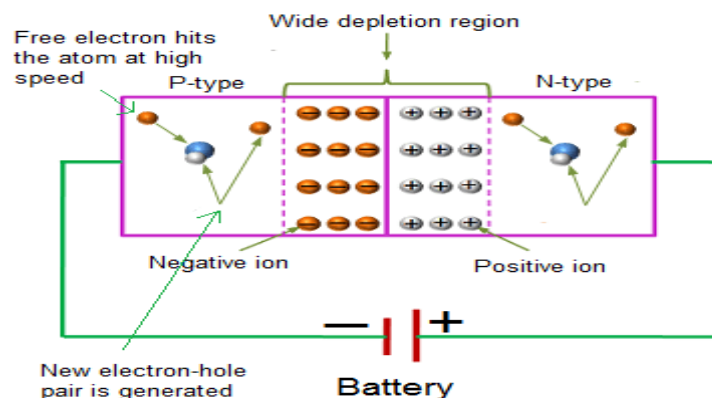
Zener diodes are heavily doped. Therefore, the width of depletion region in zener diode is very thin. Because of this thin depletion layer or region, reverse breakdown occurs at lower voltages in zener diode. On the other hand, avalanche diodes are lightly doped. Therefore, the width of depletion layer in avalanche diode is very wide compared to the zener diode. Because of this wide depletion region, reverse breakdown occurs at higher voltages in avalanche diode. The breakdown voltage of avalanche diode is carefully set by controlling the doping level during manufacture.

Avalanche diode symbol



The symbol of avalanche diode is similar to the normal diode but with the bend edges on the vertical bar.

**Working:** When reverse bias voltage is applied to the avalanche diode, the free electrons (majority carriers) in the n-type semiconductor and the holes (majority carriers) in the p-type semiconductor are moved away from the junction. As a result, the width of depletion region increases. Therefore, the majority carriers will not carry electric current. However, the minority carriers (free electrons in p-type and holes in n-type) experience a repulsive force from external voltage.



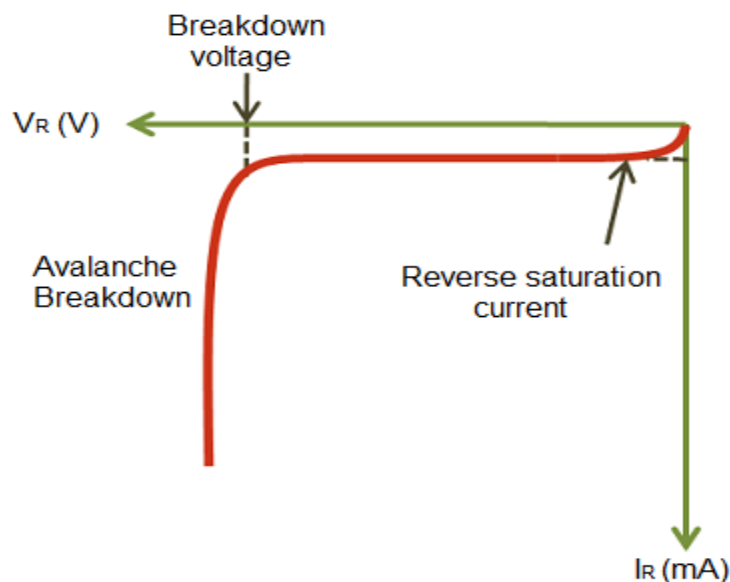
As a result, the minority carriers flow from p-type to n-type and n-type to p-type by carrying the electric current. However, electric current carried by minority carriers is very small. This small electric current carried by minority carriers is called reverse leakage current.

If the reverse bias voltage applied to the avalanche diode is further increased, the minority carriers (free electrons or holes) will gain large amount of energy and accelerated to greater

velocities. The free electrons moving at high speed will collide with the atoms and transfer their energy to the valence electrons.

The valance electrons which gains enough energy from the high-speed electrons will be detached from the parent atom and become free electrons. These free electrons are again accelerated. When these free electrons again collide with other atoms, they knock off more electrons. Because of this continuous collision with the atoms, a large number of minority carriers (free electrons or holes) are generated. These large numbers of free electrons carry excess current in the diode.

When the reverse voltage applied to the avalanche diode continuously increases, at some point the junction breakdown or avalanche breakdown occurs. At this point, a small increase in voltage will suddenly increases the electric current. This sudden increase of electric current may permanently destroys the normal p-n junction diode. However, avalanche diodes may not be destroyed because they are carefully designed to operate in avalanche breakdown region.



The breakdown voltage of the avalanche diode depends on the doping density. Increasing the doping density will decreases the breakdown voltage of the avalanche diode.

### Applications of avalanche diodes

- Avalanche diodes can be used as white noise generators.
- Avalanche diodes are used in protecting circuits.
- This makes the diode to start performing current without injuring itself, and switches the extreme power away from the electrical circuits to its ground terminal.

## Short answer questions

1. What are intrinsic semiconductors? Give examples.
2. What are extrinsic semiconductors? Give examples.
3. Distinguish between intrinsic and extrinsic semiconductors.
4. Define Fermi level and mention its position in intrinsic and extrinsic semiconductors at 0K.
5. Explain about carrier generation and recombination mechanism.
6. Write the applications of Hall effect.
7. Write short note on Zener diode.
8. Explain V-I characteristics of p-n junction.

## Essay questions

1. Derive an expression for concentration of electrons and holes in intrinsic semiconductor.
2. Derive an expression for carrier concentration in n-type and p-type semiconductor.
3. Explain drift and diffusion mechanism in semiconductors.
4. What is Hall effect? Derive expression for Hall coefficient.
5. Explain how a p-n junction is formed and explain with neat energy level diagram of p-n junction.
6. Explain construction and working of BJT.
7. Explain construction and working of avalanche diode.
8. Explain construction and working of PIN diode.
9. Explain construction and working of LED.
10. Explain construction and working of solarcell.

## Multiple answer questions

1. At 0K pure silicon is [     ]  
(a) an intrinsic semiconductor      (b) extrinsic semiconductor  
(c) insulator      (d) superconductor
2. The majority carriers in p- type Ge are [     ]  
(a) free electrons (b) holes (c) ions (d) conduction electrons
3. The electric current in a pure semiconductor when voltage is applied is due to [     ]  
(a) Electrons (b) holes (c) ions (d) holes and electrons
4. In an intrinsic semiconductor at 0 K, the Fermi level lies [     ]  
(a) at the centre of top of the valence band and bottom of the conduction band  
(b) at the centre of donor level and intrinsic Fermi level  
(c) at the centre of donor level and bottom of the conduction band  
(d) at the centre of acceptor level and top of the valence band
5. The density of charge carriers in a pure semiconductor is proportional to [     ]  
(a)  $\exp(-E_g/kT)$  (b)  $\exp(2E_g/kT)$  (c)  $\exp(-E_g/kT^2)$  (d)  $\exp(-E_g/2kT)$

## Applied Physics

## I -B.Tech

6. Life time of LEDs is about [   ]  
(a)  $10^2$  hours (b)  $10^3$  hours (c)  $10^4$  hours (d)  $10^5$  hours
7. The thickness of depletion region is in the order of [   ]  
(a)  $6 \times 10^{-3}$  cm (b)  $6 \times 10^{-9}$  cm (c)  $6 \times 10^{-6}$  cm (d)  $6 \times 10^{-12}$  cm
8. In a LED, if  $E_g$  is the band gap, the wavelength of emission is given by [   ]  
(a)  $\lambda = E_g / h\nu$  (b)  $\lambda = h\nu / E_g$  (c)  $\lambda = E_g / hc$  (d)  $\lambda = hc / E_g$
9. Response time of LEDs in the order of [   ]  
(a)  $10^{-3}$  s (b)  $10^{-6}$  s (c)  $10^{-9}$  s (d)  $10^{-12}$  s
10. Intrinsic semiconductor atom possesses--- valance electrons [   ]  
(a) 3 (b) 5 (c) 4 (d) 7
11. Which of the following a semiconductor material [   ]  
(a) Aluminium (b) Rubber (c) Silicon (d) Phosphorous
12. At 0 K pure silicon is [   ]  
(a) Insulator (b) conductor  
(c) Semiconductor (d) super conductor
13. At room temperature , semiconductor possesses—co-efficient of resistivity [   ]  
(a) Positive (b) negative (c) zero (d) neutral
14. Selenium is [   ]  
(a) Intrinsic semiconductor (b) Extrinsic semiconductor  
(c) p-type semiconductor (d) n-type semiconductor
15. Electron- hole recombination results [   ]  
(a) Release of energy (b) Absorption of energy (c) Both a&b (d) None
16. In a reverse bias diode the depletion layer is [   ]  
(a)Thick (b) thin (c) negligible (d) very large
17. In intrinsic semiconductor the carrier concentration varies [   ]  
(a)  $1/T$  (b)  $T^{3/2}$  (c)  $T^2$  (d)  $T$
18. The resistivity of pure silicon  $0^\circ\text{C}$  is  $300\Omega\text{m}$ . The intrinsic scarier concentration per cubic meter is ( $\mu_e=0.14$  and  $\mu_h=0.05\text{m}^3\text{V}^{-1}\text{s}^{-1}$ ) [   ]  
(a)  $1.095 \times 10^{17}$  (b)  $2.19 \times 10^{17}$  (c)  $10^{17}$  (d) zero
19. In conductors energy gap is [   ]  
(a) 0.6 (b) 0.7 (c) 0 (d) 1.1
20. Hall coefficient for n type material [   ]  
(a)negative (b)positive (c)neutral (d)none
21. Hall coefficient for p type material [   ]  
(a)negative (b)positive (c)neutral (d)none
22. The energy gap for Germanium----- Ev [   ]  
(a)0.75 (b)1.1 (c)1.5 (d)2

## Applied Physics

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23. The energy gap for Silicon----- eV [   ]  
 (a) 0.75 (b) 1.1 (c) 1.5 (d) 2
24. The Charge carriers in intrinsic semi conductor [   ]  
 (a) electrons (b) holes (c) both a&b (d) none
25. Majority charge carriers in P- type semiconductor [   ]  
 (a) Holes (b) Electrons  
 (c) Protons (d) Photons

### Fill in the blanks

- Solar is a ----- device
- LEDs are made up of ----- semiconductors
- PIN diode consists of.....
- Photo voltaic cell works on the principle of-----
- When reverse biased, PIN diode acts like a -----

### Choose the Correct Alternative

- Photovoltaic cell works on the principle of..... [   ]  
 a) forward bias      b) photovoltaic effect      c) reverse bias      d) none
- PIN diode consists of..... [   ]  
 a) 1 operating regions      b) 2 operating regions      c) 3 operating regions      d) none
- Photo diode is normally..... [   ]  
 a) forward biased      b) reverse biased      c) none      d) emitting light
- Photo conductivity increases with addition of [   ]  
 A) impurity      b) nickel      c) iron      d) none
- A solar cell is simply photodiode which is operated at [   ]  
 a) zero biased voltage      b) constant biased voltage      c) forward biased voltage  
 d) reverse biased voltage